BSDS – Statistical Learning I

Problem Set

Complete Detailed Solution Continuation

Questions 37–70

Full question statements, prerequisite theory, and step-by-step reasoning

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Prepared from the supplied assignment and lecture notes
(material through W15)

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How to read this solution set

This document continues the supplied `PS_midsem_Solns.pdf`, which ends at Question 36. Therefore, Questions 1–36 are not repeated here. The question statements below are transcribed from the uploaded `Problem_Set_5(28th_oct).pdf`; the theory and notation are drawn from the uploaded `until_W12(1).txt`, `W13.pdf`, `W14.pdf`, and `W15.pdf`. Each question begins with the full question, followed by the theory needed, then a numbered derivation. Short transition explanations are included so that the reason for each mathematical move is visible.

Source boundary. The supplied notes explicitly develop Bayes classification, density estimation, nearest-neighbour methods, perceptrons, logistic/probit GLMs, SVMs and kernels, bagging/random forests, and discrete AdaBoost. They do not explicitly define out-of-bag error, regression AdaBoost, Gradient Boosting, or XGBoost. Where these appear in Questions 46 and 70, the solution identifies what is directly supported by the notes and labels any concise supplementary explanation as such. No external reference or web source is used; supplementary explanations are clearly labelled.

Question 37

Whole question from the assignment.

> Consider a classification problem between two multivariate normal distributions $N(\mu_1, \Sigma)$ and $N(\mu_2, \Sigma)$. If the two classes have equal prior probabilities, show that (i) the Bayes classifier is linear and (ii) the Bayes risk is a decreasing function of $(\mu_1 - \mu_2)^\top \Sigma^{-1} (\mu_1 - \mu_2)$.

Theory required for this solution.

For two classes with prior probabilities π_1, π_2 and class-conditional densities f_1, f_2, the Bayes rule assigns x to Class 1 when

$$\pi_1 f_1(x) > \pi_2 f_2(x).$$

For a d-dimensional normal distribution,

$$f_j(x) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(x-\mu_j)^\top \Sigma^{-1}(x-\mu_j)\right].$$

When the covariance matrices and priors are equal, the normalising constants cancel in the density comparison. A linear classifier is one whose decision rule is determined by the sign of an affine function of x. We also use the fact that if $X \sim N(\mu, \Sigma)$, then $a^\top X + c$ is univariate normal with mean $a^\top \mu + c$ and variance $a^\top \Sigma a$.

Step 1. Write the Bayes comparison using the two normal densities. Since the priors are equal, $\pi_1 = \pi_2 = 1/2$, the rule is

$$f_1(x) > f_2(x).$$

The determinants and normalising constants are identical, so taking logarithms and cancelling common terms gives

$$-\frac{1}{2}(x-\mu_1)^\top \Sigma^{-1}(x-\mu_1) > -\frac{1}{2}(x-\mu_2)^\top \Sigma^{-1}(x-\mu_2).$$

Why this step leads to the next step. Taking logarithms is valid because the densities are positive. It also turns the comparison of exponentials into a comparison of quadratic forms, which can be simplified directly.

Step 2. Multiply by -2 and reverse the inequality:

$$(x-\mu_1)^\top \Sigma^{-1}(x-\mu_1) < (x-\mu_2)^\top \Sigma^{-1}(x-\mu_2).$$

Put $A = \Sigma^{-1}$. Expanding both sides,

$$\begin{aligned} &x^\top Ax - 2\mu_1^\top Ax + \mu_1^\top A\mu_1 \\ &\quad < x^\top Ax - 2\mu_2^\top Ax + \mu_2^\top A\mu_2. \end{aligned}$$

The $x^\top Ax$ terms cancel. Rearranging,

$$2x^\top A(\mu_1-\mu_2) > \mu_1^\top A\mu_1 - \mu_2^\top A\mu_2.$$

Equivalently,

$$\left(x - \frac{\mu_1+\mu_2}{2}\right)^\top \Sigma^{-1}(\mu_1-\mu_2) > 0.$$

This is a linear inequality in x. Hence the Bayes decision boundary is a hyperplane.

Step 3. Define the squared Mahalanobis separation

$$\Delta^2 = (\mu_1-\mu_2)^\top \Sigma^{-1}(\mu_1-\mu_2),$$

and the discriminant score

$$T(X) = \left(X - \frac{\mu_1+\mu_2}{2}\right)^\top \Sigma^{-1}(\mu_1-\mu_2).$$

Under Class 1, write $X = \mu_1 + \varepsilon$, where $\varepsilon \sim N(0,\Sigma)$. Therefore

$$T(X) = \frac{1}{2}\Delta^2 + \varepsilon^\top \Sigma^{-1}(\mu_1-\mu_2) \sim N\left(\frac{1}{2}\Delta^2, \Delta^2\right).$$

Under Class 2, the same calculation gives

$$T(X) \mid C_2 \sim N\left(-\frac{1}{2}\Delta^2, \Delta^2\right).$$

Why this step leads to the next step. We have converted a d-dimensional classification error into two one-dimensional normal tail probabilities. That makes the Bayes risk directly computable.

Step 4. The Bayes rule assigns Class 1 when $T(X) > 0$. Thus

$$\mathbb{P}(\text{error} \mid C_1) = \mathbb{P}(T \le 0 \mid C_1) = \Phi\left(-\frac{\Delta}{2}\right),$$

and, by symmetry,

$$\mathbb{P}(\text{error} \mid C_2) = \mathbb{P}(T > 0 \mid C_2) = \Phi\left(-\frac{\Delta}{2}\right).$$

With equal priors,

$$R^* = \frac{1}{2}\mathbb{P}(\text{error} \mid C_1) + \frac{1}{2}\mathbb{P}(\text{error} \mid C_2) = \Phi\left(-\frac{\Delta}{2}\right).$$

The standard normal cdf is increasing, while $-\Delta/2$ decreases as Δ^2 increases. Therefore R^* is decreasing in Δ^2.

Final conclusion.

The Bayes classifier is the linear rule

$$\boxed{\delta(x) = 1 \quad \text{if} \quad \left(x - \frac{\mu_1 + \mu_2}{2}\right)^\top \Sigma^{-1}(\mu_1 - \mu_2) > 0,}$$

and its Bayes risk is

$$\boxed{R^* = \Phi\left(-\frac{1}{2}\sqrt{(\mu_1 - \mu_2)^\top \Sigma^{-1}(\mu_1 - \mu_2)}\right).}$$

It decreases as the squared Mahalanobis separation increases.

Question 38

Whole question from the assignment.

Consider a classification problem with equal priors, where the two distributions have densities

$$f_1(x) = C|\Sigma|^{-1/2}\psi\Big((x - \mu_1)^\top \Sigma^{-1}(x - \mu_1)\Big)$$

and

$$f_2(x) = C|\Sigma|^{-1/2}\psi\Big((x - \mu_2)^\top \Sigma^{-1}(x - \mu_2)\Big),$$

respectively. If $\psi : \mathbb{R}_+ \to \mathbb{R}_+$ is a strictly decreasing function, show that the Bayes classifier is linear. Will the Bayes classifier remain linear if the priors are not equal? Justify your answer.

Theory required for this solution.

The Bayes rule compares $\pi_1 f_1(x)$ and $\pi_2 f_2(x)$. Strict decrease of ψ reverses comparisons of its arguments:

$$\psi(u) > \psi(v) \quad \Longleftrightarrow \quad u < v.$$

The common factor $C|\Sigma|^{-1/2}$ does not affect which class has the larger weighted density. A difference of two quadratic forms with the same matrix Σ^{-1} and different centres is an affine function of x because the common $x^\top \Sigma^{-1} x$ term cancels.

Step 1. With equal priors, compare $f_1(x)$ and $f_2(x)$. The common positive factor cancels, so

$$\psi(q_1(x)) > \psi(q_2(x)),$$

where

$$q_j(x) = (x - \mu_j)^\top \Sigma^{-1}(x - \mu_j).$$

Since ψ is strictly decreasing, the comparison is equivalent to

$$q_1(x) < q_2(x).$$

Step 2. Expand this inequality. Let $A = \Sigma^{-1}$. Then

$$\begin{aligned} q_2(x) - q_1(x) &= (x - \mu_2)^\top A(x - \mu_2) - (x - \mu_1)^\top A(x - \mu_1) \\ &= 2x^\top A(\mu_1 - \mu_2) + \mu_2^\top A\mu_2 - \mu_1^\top A\mu_1. \end{aligned}$$

The right-hand side is affine in x. Thus the decision boundary $q_1(x) = q_2(x)$ is a hyperplane, and the Bayes classifier is linear.

Step 3. Now suppose $\pi_1 \neq \pi_2$. The Bayes comparison becomes

$$\pi_1\psi(q_1(x)) > \pi_2\psi(q_2(x)),$$

or, equivalently,

$$\frac{\psi(q_1(x))}{\psi(q_2(x))} > \frac{\pi_2}{\pi_1}.$$

Strict monotonicity alone does not imply that this ratio inequality is an affine inequality in x. Therefore a linear boundary is not guaranteed.

Step 4. To see why, choose a valid strictly decreasing function

$$\psi(t) = e^{-t^2}, \qquad \Sigma = I_2, \qquad \mu_1 = (a, 0)^\top, \quad \mu_2 = (-a, 0)^\top,$$

where $a > 0$. Put $q_1 = (x_1 - a)^2 + x_2^2$ and $q_2 = (x_1 + a)^2 + x_2^2$. The log weighted-density ratio is

$$\begin{aligned}\log \frac{\pi_1 f_1(x)}{\pi_2 f_2(x)} &= \log \frac{\pi_1}{\pi_2} - q_1^2 + q_2^2 \\ &= \log \frac{\pi_1}{\pi_2} + (q_2 - q_1)(q_2 + q_1) \\ &= \log \frac{\pi_1}{\pi_2} + 8ax_1(x_1^2 + x_2^2 + a^2).\end{aligned}$$

For unequal priors, the boundary is a nonzero level set of the cubic expression $x_1(x_1^2 + x_2^2 + a^2)$, which is not a hyperplane in general.

Final conclusion.

For equal priors, the Bayes classifier is linear because strict decrease converts the density comparison into a comparison of two common-covariance Mahalanobis distances. For unequal priors, linearity is *not guaranteed* under an arbitrary strictly decreasing ψ. Special cases, such as the Gaussian radial form $\psi(t) = e^{-t/2}$, can still give a linear boundary, but this does not follow from monotonicity alone.

Question 39

Whole question from the assignment.

> In a binary classification problem between two d-dimensional normal distributions $N(\mu, \Sigma)$ and $N(\mu, 4\Sigma)$, show that the Bayes risk can be expressed in the form
>
> $$a_0 + a_1 G_d(b_1) + a_2 G_d(b_2),$$
>
> where $G_d(\cdot)$ is the cdf of the chi-square distribution with d degrees of freedom. Compute a_0, a_1, a_2, b_1 and b_2 in terms of prior probabilities and parameters of the two classes.

Theory required for this solution.

The Bayes classifier chooses the class with the larger prior-weighted density. If $Z \sim N_d(0, I_d)$, then $Z^\top Z \sim \chi^2_d$. Also, if $X \sim N_d(\mu, 4\Sigma)$, then the squared Mahalanobis radius based on Σ is

four times a χ^2_d variable. Denote the chi-square cdf by

$$G_d(t) = \mathbb{P}(\chi^2_d \le t),$$

and set $G_d(t) = 0$ for $t \le 0$.

Step 1. Let

$$Q(x) = (x-\mu)^\top \Sigma^{-1}(x-\mu).$$

The two densities are

$$\begin{aligned} f_1(x) &= (2\pi)^{-d/2}|\Sigma|^{-1/2}e^{-Q(x)/2}, \\ f_2(x) &= (2\pi)^{-d/2}|4\Sigma|^{-1/2}e^{-Q(x)/8} \\ &= (2\pi)^{-d/2}|\Sigma|^{-1/2}2^{-d}e^{-Q(x)/8}, \end{aligned}$$

because $|4\Sigma| = 4^d|\Sigma|$.

Step 2. Classify to Class 1 when $\pi_1 f_1(x) > \pi_2 f_2(x)$. Cancelling common constants and taking logarithms gives

$$\log \pi_1 - \frac{Q(x)}{2} > \log \pi_2 - d\log 2 - \frac{Q(x)}{8}.$$

Hence

$$Q(x) < \frac{8}{3}\left\{\log\frac{\pi_1}{\pi_2} + d\log 2\right\} =: b.$$

Thus the Bayes rule selects Class 1 inside the ellipsoid $Q(x) < b$, and Class 2 outside it. If $b \le 0$, the ellipsoid is empty up to a probability-zero boundary, so the rule selects Class 2 almost surely.

Why this step leads to the next step. The decision is now expressed through one scalar random variable, $Q(X)$. The next step is to use its distribution under each class to calculate the two types of error.

Step 3. Under Class 1, write $X = \mu + \Sigma^{1/2}Z$ with $Z \sim N_d(0, I_d)$. Then

$$Q(X) = Z^\top Z \sim \chi^2_d.$$

Under Class 2, write $X = \mu + 2\Sigma^{1/2}Z$; hence

$$Q(X) = 4Z^\top Z \quad \text{and so} \quad Q(X)/4 \sim \chi^2_d.$$

Step 4. Class 1 is misclassified when $Q(X) \ge b$. Therefore

$$\mathbb{P}(\text{error} \mid C_1) = 1 - G_d(b).$$

Class 2 is misclassified when $Q(X) < b$. Therefore

$$\mathbb{P}(\text{error} \mid C_2) = G_d(b/4).$$

Weighting these two conditional error probabilities by their priors,

$$R^* = \pi_1\{1 - G_d(b)\} + \pi_2 G_d(b/4).$$

This expression also covers $b \le 0$ under the convention $G_d(t) = 0$ for non-positive t.

Final conclusion.
The requested coefficients are

$$\boxed{a_0 = \pi_1, \quad a_1 = -\pi_1, \quad a_2 = \pi_2,}$$

with

$$\boxed{b_1 = \frac{8}{3}\left(\log\frac{\pi_1}{\pi_2} + d\log 2\right), \qquad b_2 = \frac{b_1}{4}.}$$

Thus

$$\boxed{R^* = \pi_1 - \pi_1 G_d(b_1) + \pi_2 G_d(b_2).}$$

Question 40

Whole question from the assignment.

Show that for a classification problem with equal prior probabilities, the Bayes risk is $1/2$ if and only if the two distributions are identical.

Theory required for this solution.
For two classes with equal priors, the Bayes classifier chooses the larger of $f_1(x)$ and $f_2(x)$. Its risk is the integral of the smaller prior-weighted density:

$$R^* = \frac{1}{2}\int \min\{f_1(x), f_2(x)\}\,dx.$$

For any two densities, $\int f_1 = \int f_2 = 1$ and

$$\min(f_1, f_2) = \frac{1}{2}(f_1 + f_2 - |f_1 - f_2|).$$

Step 1. Substitute the identity for the minimum into the Bayes risk:

$$\begin{aligned} R^* &= \frac{1}{2}\int \frac{1}{2}\{f_1(x) + f_2(x) - |f_1(x) - f_2(x)|\}\,dx \\ &= \frac{1}{4}\left\{1 + 1 - \int |f_1(x) - f_2(x)|\,dx\right\} \\ &= \frac{1}{2} - \frac{1}{4}\int |f_1(x) - f_2(x)|\,dx. \end{aligned}$$

Step 2. Since the integral of an absolute value is non-negative,

$$R^* \le \frac{1}{2}.$$

Equality holds if and only if

$$\int |f_1(x) - f_2(x)|\,dx = 0.$$

A non-negative function has integral zero exactly when it is zero almost everywhere. Thus equality is equivalent to

$$f_1(x) = f_2(x) \quad \text{for almost every } x,$$

which means the two distributions are identical.

Step 3. Conversely, if the distributions are identical, then $f_1 = f_2$ almost everywhere. The classifier has no information with which to distinguish the labels, and the Bayes risk is

$$R^* = \frac{1}{2}\int f_1(x)\,dx = \frac{1}{2}.$$

Final conclusion.

$$\boxed{R^* = \frac{1}{2} \quad \Longleftrightarrow \quad f_1 = f_2 \text{ almost everywhere.}}$$

For equal priors, the Bayes risk equals one-half exactly when the two class distributions are identical.

Question 41

Whole question from the assignment.

> Prove or disprove the following statement: "In a classification problem with equal prior probabilities, if the two multivariate distributions are different (even if they do not differ in their locations), there must be a linear classifier which has misclassification probability smaller than 0.5".

Theory required for this solution.

A linear classifier with an intercept can assign one class to a half-space

$$A = \{x : a^\top x \le t\}$$

and the other class to its complement. A multivariate distribution is determined by the distributions of all its one-dimensional linear projections. To see the idea, the characteristic function of $a^\top X$ at argument t is the joint characteristic function of X evaluated at ta. If every projection had the same distribution under the two classes, their joint characteristic functions would agree at every vector, and hence the joint distributions would be identical. This is the projection-characterisation principle for multivariate distributions.

Step 1. Since the two multivariate distributions are different, there exists a nonzero direction a for which the projected random variables

$$U_1 = a^\top X \mid C_1, \qquad U_2 = a^\top X \mid C_2$$

have different distributions.

Step 2. Different one-dimensional distributions must have different cdfs at some threshold t. Thus, for some t,

$$\mathbb{P}(U_1 \le t) \ne \mathbb{P}(U_2 \le t).$$

If the inequality is in the wrong direction, interchange the class labels or use the complementary half-space. We may therefore choose a, t so that

$$\mathbb{P}(X \in A \mid C_1) > \mathbb{P}(X \in A \mid C_2), \qquad A = \{x : a^\top x \le t\}.$$

Step 3. Define the affine linear classifier to assign Class 1 on A and Class 2 on A^c. With equal priors, its risk is

$$\begin{aligned} R &= \frac{1}{2}\mathbb{P}(X \notin A \mid C_1) + \frac{1}{2}\mathbb{P}(X \in A \mid C_2) \\ &= \frac{1}{2}\left[1 - \mathbb{P}(X \in A \mid C_1) + \mathbb{P}(X \in A \mid C_2)\right] \\ &< \frac{1}{2}. \end{aligned}$$

Why this step leads to the next step. The strict inequality follows from the choice of a half-space on which the first distribution places more probability than the second. The intercept t is important: the separating hyperplane need not pass through the origin.

Final conclusion.

The statement is **true** when "linear classifier" includes an affine hyperplane with an intercept. If the phrase were restricted to hyperplanes passing through the origin, the result would not follow; that is a narrower claim of the kind distinguished in Question 36.

Question 42

Whole question from the assignment.

> If $X_1, X_2, \ldots, X_d$ are i.i.d. $N(0,1)$, $S = \sum_{i=1}^{d} X_i^2$ follows the chi-square distribution with d degrees of freedom. Using this result, find the volume of the d-dimensional unit ball and the surface area of the d-dimensional unit sphere.

Theory required for this solution.

For $X = (X_1, \ldots, X_d)^\top \sim N_d(0, I_d)$, the squared radius

$$S = \|X\|^2 = \sum_{i=1}^{d} X_i^2 \sim \chi^2_d.$$

The chi-square density is

$$g_d(s) = \frac{s^{d/2-1}e^{-s/2}}{2^{d/2}\Gamma(d/2)}, \qquad s > 0.$$

Let A_d denote the surface area of the unit sphere in $\mathbb{R}^d$, and V_d the volume of the unit ball. Polar integration gives a radial density proportional to $A_d r^{d-1}$.

Step 1. Let $R = \|X\|$. Since $S = R^2$, the transformation $s = r^2$ gives

$$f_R(r) = 2r\, g_d(r^2) = \frac{2^{1-d/2}}{\Gamma(d/2)} r^{d-1} e^{-r^2/2}, \qquad r > 0.$$

Step 2. On the other hand, the standard normal density in $\mathbb{R}^d$ is

$$(2\pi)^{-d/2} e^{-\|x\|^2/2}.$$

Integrating over a thin spherical shell of radius r and thickness dr, whose volume is approximately $A_d r^{d-1} dr$, gives

$$f_R(r) = A_d (2\pi)^{-d/2} r^{d-1} e^{-r^2/2}.$$

Step 3. Equate the two expressions for $f_R(r)$ and cancel the common positive factor $r^{d-1}e^{-r^2/2}$:

$$A_d(2\pi)^{-d/2} = \frac{2^{1-d/2}}{\Gamma(d/2)}.$$

Therefore

$$A_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}.$$

Step 4. The volume of a radius-r ball is $V_d r^d$, and differentiating with respect to r at $r = 1$ shows that its derivative equals the surface area A_d. Consequently $A_d = dV_d$, so

$$V_d = \frac{A_d}{d} = \frac{\pi^{d/2}}{\Gamma(d/2+1)},$$

where we used $\Gamma(z+1) = z\Gamma(z)$.

Final conclusion.

$$\boxed{\mathrm{Vol}(B_d(1)) = \frac{\pi^{d/2}}{\Gamma(d/2+1)}} \quad \text{and} \quad \boxed{\mathrm{Area}(S^{d-1}) = \frac{2\pi^{d/2}}{\Gamma(d/2)}}.$$

Question 43

Whole question from the assignment.

> Let $\widehat{f}_h$ be the histogram estimator of a density function based on a common bin-width h. If the underlying density function f has a bounded derivative, and $h = h_n = O(n^{-\delta})$ for some $\delta \in (0,1)$ (so that $h \to 0$ and $nh \to \infty$), show that for any fixed x, $\widehat{f}_h(x)$ converges to $f(x)$ in probability as the sample size n grows to infinity.

Theory required for this solution.

For a bin $B_n(x)$ of width h_n containing x, let $N_n(x)$ be the number of observations in that bin. The histogram estimate is

$$\widehat{f}_h(x) = \frac{N_n(x)}{nh_n}.$$

If the observations are i.i.d. from density f, then $N_n(x) \sim \mathrm{Bin}(n, p_n)$, where $p_n = \int_{B_n(x)} f(u)\,du$. A bounded derivative means that f is Lipschitz: for some constant L, $|f(u) - f(x)| \le L|u - x|$.

Step 1. Compute the expectation:

$$\mathbb{E}\{\widehat{f}_h(x)\} = \frac{1}{nh_n}\mathbb{E}\{N_n(x)\} = \frac{p_n}{h_n} = \frac{1}{h_n}\int_{B_n(x)} f(u)\,du.$$

This is the average value of the density over the bin containing x.

Step 2. Bound the bias. Every u in the bin is within distance at most h_n of x. Therefore

$$\begin{aligned}\left|\mathbb{E}\{\widehat{f}_h(x)\} - f(x)\right| &= \left|\frac{1}{h_n}\int_{B_n(x)} \{f(u) - f(x)\}\,du\right| \\ &\le \frac{1}{h_n}\int_{B_n(x)} L|u - x|\,du \\ &\le Lh_n \longrightarrow 0.\end{aligned}$$

Thus the bias tends to zero.

Step 3. Compute and bound the variance:

$$\begin{aligned}\operatorname{Var}\{\widehat{f}_h(x)\} &= \frac{1}{n^2 h_n^2}\operatorname{Var}\{N_n(x)\} \\ &= \frac{p_n(1-p_n)}{nh_n^2} \\ &\le \frac{p_n}{nh_n^2}.\end{aligned}$$

If f is bounded near the fixed point x, say by M, then $p_n \le Mh_n$, and hence

$$\operatorname{Var}\{\widehat{f}_h(x)\} \le \frac{M}{nh_n} \longrightarrow 0.$$

Step 4. The mean squared error is the variance plus squared bias:

$$\mathbb{E}\left[\{\widehat{f}_h(x) - f(x)\}^2\right] = \operatorname{Var}\{\widehat{f}_h(x)\} + \left(\mathbb{E}\widehat{f}_h(x) - f(x)\right)^2 \longrightarrow 0.$$

By Markov's inequality applied to the squared error, for every $\varepsilon > 0$,

$$\mathbb{P}\left(|\widehat{f}_h(x) - f(x)| > \varepsilon\right) \le \frac{\mathbb{E}[(\widehat{f}_h(x) - f(x))^2]}{\varepsilon^2} \longrightarrow 0.$$

The assignment explicitly states both $h_n \to 0$ and $nh_n \to \infty$. Those two conditions are what the proof uses. Strictly speaking, the upper bound $h_n = O(n^{-\delta})$ by itself guarantees $h_n \to 0$ but does not, without a corresponding lower-rate condition, guarantee $nh_n \to \infty$. The parenthetical conditions in the question must therefore be read as additional assumptions.

Final conclusion.
For every fixed x,

$$\boxed{\widehat{f}_{h_n}(x) \xrightarrow{P} f(x)}$$

because the histogram bias tends to zero and its variance also tends to zero under $h_n \to 0$ and $nh_n \to \infty$.

Question 44

Whole question from the assignment.

> Let $X_1, X_2, \ldots, X_n$ be independent copies of a random variable X, which follows a distribution with density f. If $\widehat{f}_h$ is the kernel density estimator of f based on a symmetric kernel with bandwidth h, where f is bounded and has a bounded, continuous second derivative and the kernel is a bounded symmetric density with finite second moment, show that for any given x,
>
> $$\operatorname{Bias}(\widehat{f}_h(x)) = O(h^2) \quad \text{and} \quad \operatorname{Var}(\widehat{f}_h(x)) = O\left(\frac{1}{nh}\right)$$
>
> as $h \to 0$.

Theory required for this solution.
The one-dimensional kernel density estimator is

$$\widehat{f}_h(x) = \frac{1}{nh}\sum_{i=1}^{n} K\Big(\frac{x - X_i}{h}\Big),$$

where K is a density: $\int K(u)du = 1$. Symmetry gives $K(u) = K(-u)$ and therefore $\int uK(u)du = 0$. The finite second moment means $\int u^2K(u)du < \infty$. Independence of the observations lets the variance of the average be reduced by a factor n.

Step 1. Take expectation and make the substitution $u = (x - t)/h$:

$$\begin{aligned}\mathbb{E}\{\widehat{f}_h(x)\} &= \frac{1}{h}\mathbb{E}\left[K\Big(\frac{x - X}{h}\Big)\right] \\ &= \frac{1}{h}\int K\Big(\frac{x - t}{h}\Big) f(t)\, dt \\ &= \int K(u) f(x - hu)\, du.\end{aligned}$$

Step 2. Apply the second-order Taylor expansion around x:

$$f(x - hu) = f(x) - huf'(x) + \frac{h^2u^2}{2} f''(x - \theta hu),$$

for some $\theta \in (0, 1)$. Substituting into the expectation,

$$\begin{aligned}\mathbb{E}\widehat{f}_h(x) = f(x)\int K(u)du - hf'(x)\int uK(u)du \\ + \frac{h^2}{2}\int u^2K(u)f''(x - \theta hu)du.\end{aligned}$$

The first integral is 1 and the second is 0 by symmetry. If $|f''| \le M_2$, then

$$|\mathbb{E}\widehat{f}_h(x) - f(x)| \le \frac{h^2M_2}{2}\int u^2K(u)du = O(h^2).$$

Step 3. For the variance, independence gives

$$\mathrm{Var}\{\widehat{f}_h(x)\} = \frac{1}{n}\mathrm{Var}\left\{\frac{1}{h}K\Big(\frac{x - X}{h}\Big)\right\} \le \frac{1}{nh^2}\mathbb{E}\left[K^2\Big(\frac{x - X}{h}\Big)\right].$$

Step 4. Use the boundedness of f and substitute $u = (x - t)/h$:

$$\begin{aligned}\mathbb{E}\left[K^2\Big(\frac{x - X}{h}\Big)\right] &= \int K^2\Big(\frac{x - t}{h}\Big) f(t)dt \\ &= h\int K^2(u)f(x - hu)du \\ &\le h\|f\|_\infty \int K^2(u)du.\end{aligned}$$

Since K is bounded and has finite second moment as a density, $\int K^2 < \infty$. Therefore

$$\mathrm{Var}\{\widehat{f}_h(x)\} \le \frac{\|f\|_\infty}{nh}\int K^2(u)du = O\Big(\frac{1}{nh}\Big).$$

Final conclusion.

$$\boxed{\text{Bias}(\widehat{f}_h(x)) = O(h^2), \qquad \text{Var}(\widehat{f}_h(x)) = O\left(\frac{1}{nh}\right).}$$

Symmetry removes the first-order Taylor term; boundedness controls the variance after the change of variables.

Question 45

Whole question from the assignment.

> Consider a binary classification problem with π and $1-\pi$ as the prior probabilities of the two classes and we have 10 bivariate observations from each of these classes. Let $\delta_h(\cdot)$ be the kernel discriminant analysis rule constructed based on the Gaussian kernel with bandwidth h. Draw a rough sketch of the class boundary when (i) $\pi = 0.4$, $h \downarrow 0$, (ii) $\pi = 0.4$, $h \uparrow \infty$, (iii) $\pi = 0.5$, $h \downarrow 0$, (iv) $\pi = 0.5$, $h \uparrow \infty$.

Theory required for this solution.

Kernel discriminant analysis (KDA) classifies x by comparing

$$\pi_1 \widehat{f}_{1,h}(x) \quad \text{and} \quad \pi_2 \widehat{f}_{2,h}(x).$$

For a Gaussian kernel and a fixed finite sample, when h is very small, the nearest training observation contributes exponentially more than the other observations. Thus KDA behaves approximately like the 1-nearest-neighbour classifier; its boundary is a collection of Voronoi boundaries. For very large h, the density estimates become nearly flat. If priors differ, the larger prior dominates. With equal priors, the first nonconstant term in the large-bandwidth expansion gives a linear boundary.

Step 1. Case (i): $\pi_1 = 0.4$ and $\pi_2 = 0.6$, with $h \downarrow 0$. There are 10 observations in each class. Away from ties, the Gaussian kernel contribution from the nearest point dominates all other kernel contributions. The prior is only a fixed multiplicative factor, whereas the difference in squared distances is magnified by $1/h^2$. Therefore the rule is approximately 1-NN. Draw Voronoi cells around all 20 observations and label each cell by the class of its nearest training point.

Step 2. Case (ii): $\pi_1 = 0.4$ and $\pi_2 = 0.6$, with $h \uparrow \infty$. For large h, the two estimated densities become nearly equal and nearly constant at a fixed x. The prior weights then decide the comparison:

$$0.4\widehat{f}_{1,h}(x) < 0.6\widehat{f}_{2,h}(x)$$

for sufficiently large h. Thus the entire plane is assigned to Class 2. There is no finite separating boundary in this limiting sketch.

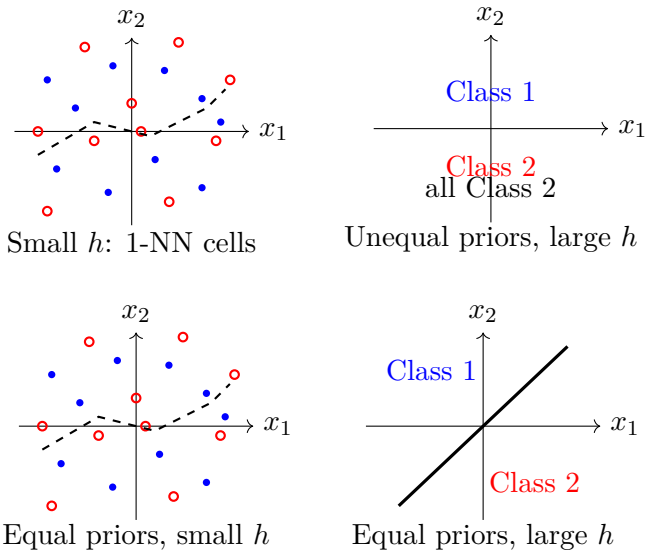
Step 3. Case (iii): $\pi_1 = \pi_2 = 0.5$ and $h \downarrow 0$. The same nearest-observation dominance as in Step 1 applies. The rough boundary is again the Voronoi-type 1-NN boundary. Since the class sample sizes are equal, changing the prior from 0.4/0.6 to 0.5/0.5 does not change which observation is uniquely closest in the zero-bandwidth limit; it can matter near distance ties for a finite, nonzero bandwidth.

Step 4. Case (iv): $\pi_1 = \pi_2 = 0.5$ and $h \uparrow \infty$. The leading prior term cancels. The next term depends on expected squared distances to the class distributions. Let μ_j and Σ_j denote the mean

vector and dispersion matrix of Class j. The large-h boundary is approximated by

$$\|x-\mu_2\|^2+\operatorname{tr}(\Sigma_2)=\|x-\mu_1\|^2+\operatorname{tr}(\Sigma_1).$$

After expansion, the $\|x\|^2$ terms cancel, so this is a linear equation in x. Its orientation and location cannot be numerically determined because the 20 observation coordinates are not supplied in the question.



The plots are schematic only. In the last panel, the actual line depends on the class sample means and dispersion traces; no sample coordinates were supplied.

Final conclusion.

(i) $\pi=0.4$, small h: approximately a 1-NN/Voronoi boundary.

(ii) $\pi=0.4$, large h: all observations classified to Class 2, the class with prior 0.6.

(iii) $\pi=0.5$, small h: approximately a 1-NN/Voronoi boundary.

(iv) $\pi=0.5$, large h: a linear boundary determined by the difference in class mean vectors and dispersion traces.

Question 46

Whole question from the assignment.

> Describe what is the difference between bagged regression/classification and a random forest method. What is an out-of-bag estimate of error.

Theory required for this solution.

Bagging (bootstrap aggregation) repeatedly draws samples of size n with replacement, fits a tree to each sample, and aggregates the fitted trees. For regression, the aggregate is the average prediction; for classification, it is a vote. A random forest makes the additional change that, at each split, only a randomly chosen subset of predictors is considered. This makes the trees less similar to one another and helps reduce the correlation among their errors.

Step 1. Describe bagging. For $b = 1, \ldots, B$, draw a bootstrap sample of size n from the training data, fit an unpruned or suitably grown tree $\widehat{f}_b$, and store its prediction. For regression,

$$\widehat{f}_{\text{bag}}(x) = \frac{1}{B} \sum_{b=1}^{B} \widehat{f}_b(x).$$

For classification, assign the class receiving the largest number of votes among the B trees.

Step 2. Describe random forests. The bootstrap sampling and aggregation steps remain the same, but at each node only a randomly selected subset of the available predictor variables is eligible for the split. The procedure is repeated to construct B trees. Regression predictions are averaged and classification predictions are combined by majority vote.

Step 3. State the key difference. Bagging randomizes the observations through bootstrap resampling but ordinarily allows every predictor to compete at every split. A random forest adds random feature selection at each split. This decorrelates the individual trees; averaging many less-correlated trees can reduce ensemble variance.

Step 4. Define out-of-bag (OOB) prediction. An observation is out of bag for a particular tree if it was not selected in that tree's bootstrap sample. For each training observation i, collect predictions only from trees for which observation i was out of bag. Aggregate those predictions and compare with its known response/class label. The resulting regression MSE or classification misclassification proportion is the OOB error estimate.

Step 5. Explain why there are usually enough OOB trees. In a bootstrap sample of size n, the probability that a particular observation is never selected is

$$\left(1 - \frac{1}{n}\right)^n \longrightarrow e^{-1} \approx 0.368.$$

Thus, on average, about 36.8% of the original observations are left out of any one bootstrap sample and can be used as OOB observations for that tree.

The uploaded notes give the bootstrap construction, random feature-subset modification, and aggregation rules, but they do not explicitly name or define OOB error. Step 4 is the direct operational consequence of the bootstrap samples: use only trees whose bootstrap samples omitted the observation being evaluated.

Final conclusion.

Bagging = bootstrap samples + trees using all predictors at each split + averaging/voting. Random forest = bagging + random predictor subsets at each split. OOB error estimates prediction error by evaluating each training observation using only trees whose bootstrap samples did not contain it.

Question 47

Whole question from the assignment.

> Let 0_d be the origin in the d-dimensional space, and let D_n be the distance between the origin and its nearest neighbor in a set of n independent observations generated from the d-variate standard normal distribution. Show that (i) for any fixed $\epsilon > 0$, $\mathbb{P}(D_n > \epsilon) \to 0$ if d is fixed and n tends to infinity; (ii) for any fixed $M < \infty$, $\mathbb{P}(D_n > M) \to 1$ if n is fixed and d tends to infinity.

Theory required for this solution.
If $X_1, \dots, X_n$ are i.i.d. $N_d(0, I_d)$, then

$$D_n = \min_{1 \le i \le n} \|X_i\|.$$

For fixed d, every ball around the origin with positive radius has positive standard-normal probability. For increasing d, the squared norm of one standard-normal vector is χ^2_d, whose mean and variance are d and $2d$.

Step 1. Fix d and $\epsilon > 0$. Let

$$p_\epsilon = \mathbb{P}(\|X_1\| \le \epsilon).$$

The standard normal density is strictly positive everywhere, so $p_\epsilon > 0$. The event $D_n > \epsilon$ means that none of the n observations falls inside the ball of radius ϵ:

$$\mathbb{P}(D_n > \epsilon) = (1 - p_\epsilon)^n \longrightarrow 0 \qquad (n \to \infty).$$

This proves (i).

Step 2. Now fix n and let $d \to \infty$. For one observation,

$$\|X_i\|^2 \sim \chi^2_d, \qquad \mathbb{E}\|X_i\|^2 = d, \qquad \mathrm{Var}(\|X_i\|^2) = 2d.$$

For fixed M, once $d > M^2$, Chebyshev's inequality gives

$$\begin{aligned} \mathbb{P}(\|X_i\| \le M) &= \mathbb{P}(\|X_i\|^2 \le M^2) \\ &\le \mathbb{P}\left(|\|X_i\|^2 - d| \ge d - M^2\right) \\ &\le \frac{2d}{(d - M^2)^2} \longrightarrow 0. \end{aligned}$$

Step 3. Use the union bound over the fixed number n of observations:

$$\begin{aligned} \mathbb{P}(D_n \le M) &= \mathbb{P}\left(\bigcup_{i=1}^{n} \{\|X_i\| \le M\}\right) \\ &\le \sum_{i=1}^{n} \mathbb{P}(\|X_i\| \le M) \\ &= n\mathbb{P}(\|X_1\| \le M) \longrightarrow 0. \end{aligned}$$

Consequently,

$$\mathbb{P}(D_n > M) = 1 - \mathbb{P}(D_n \le M) \longrightarrow 1.$$

Final conclusion.
For fixed dimension, an increasingly large sample eventually places an observation arbitrarily close to the origin with probability tending to one. For fixed sample size, the nearest observation moves arbitrarily far from the origin in probability as dimension increases:

$$\boxed{\mathbb{P}(D_n > \epsilon) \to 0 \quad (d \text{ fixed},\ n \to \infty),}$$

$$\boxed{\mathbb{P}(D_n > M) \to 1 \quad (n \text{ fixed},\ d \to \infty).}$$

Question 48

Whole question from the assignment.

> Consider a classification problem between two distributions, one of which is uniform on $\{x \in \mathbb{R}^d : \|x\| \le a\}$ and the other one is uniform on $\{x \in \mathbb{R}^d : \|x\| \le b\}$. If $a \neq b$, show that for any fixed prior probabilities of the two classes, the error rate of the Bayes classifier converges to 0 as d grows to infinity.

Theory required for this solution.

The density of the uniform distribution on a ball of radius r is constant on that ball and zero outside it. The volume of a d-dimensional radius-r ball is $V_d r^d$. The Bayes classifier compares prior-weighted densities. Assume the two fixed priors are strictly positive; a zero prior makes the result trivial.

Step 1. Without loss of generality, suppose $0 < a < b$. Write $B_a = \{x : \|x\| \le a\}$ and $B_b = \{x : \|x\| \le b\}$. The two densities are

$$f_a(x) = \frac{1}{V_d a^d} \mathbf{1}_{B_a}(x), \qquad f_b(x) = \frac{1}{V_d b^d} \mathbf{1}_{B_b}(x).$$

Step 2. If $x \in B_b \setminus B_a$, then $f_a(x) = 0$ and $f_b(x) > 0$. The Bayes rule must classify such a point to the larger-ball class. If $x \in B_a$, compare

$$\frac{\pi_a}{V_d a^d} \quad \text{and} \quad \frac{\pi_b}{V_d b^d}.$$

The ratio of the first quantity to the second is

$$\frac{\pi_a}{\pi_b} \left(\frac{b}{a}\right)^d \longrightarrow \infty,$$

because $b/a > 1$. Thus, for all sufficiently large d, the Bayes rule classifies every point in B_a to the smaller-ball class.

Why this step leads to the next step. The limiting Bayes rule separates the two supports by radius: inside the smaller ball it chooses the smaller-ball distribution, and in the annular part of the larger ball it chooses the larger-ball distribution.

Step 3. Under the smaller-ball class, this rule makes no error. Under the larger-ball class, an error occurs only if the observation falls inside B_a. Since the class is uniform on B_b,

$$\mathbb{P}(X \in B_a \mid C_b) = \frac{\mathrm{Vol}(B_a)}{\mathrm{Vol}(B_b)} = \frac{V_d a^d}{V_d b^d} = \left(\frac{a}{b}\right)^d \longrightarrow 0.$$

Therefore the Bayes risk is

$$R_d^* = \pi_a \cdot 0 + \pi_b \left(\frac{a}{b}\right)^d \longrightarrow 0.$$

Final conclusion.

$$\boxed{R_d^* \longrightarrow 0 \quad \text{as } d \to \infty.}$$

The exponential volume ratio $(a/b)^d$ dominates every fixed positive prior ratio.

Question 49

Whole question from the assignment.

> Show that the integral of the k-nearest-neighbor density estimate is not finite in one dimension. Describe how one gets the nearest-neighbor classifier from the idea of nearest-neighbor density estimation.

Theory required for this solution.

The nearest-neighbor density estimator chooses the smallest ball around x containing its k nearest observations. In dimension d, its form is

$$\widehat{f}_k(x) = \frac{k/n}{\mathrm{Vol}(B_{r_k(x)}(x))}.$$

In one dimension, the ball is an interval of length $2r_k(x)$, so

$$\widehat{f}_k(x) = \frac{k/n}{2r_k(x)}.$$

A Bayes classifier can be constructed by estimating each class density and comparing the prior-weighted estimates.

Step 1. Sort the one-dimensional observations:

$$X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}.$$

For a query point $x \ge X_{(n)}$, the k nearest observations are the k largest sample observations. The kth nearest one is $X_{(n-k+1)}$. Hence

$$r_k(x) = x - X_{(n-k+1)},$$

and

$$\widehat{f}_k(x) = \frac{k/n}{2\{x - X_{(n-k+1)}\}}.$$

Step 2. Integrate only over the right tail. Since the estimator is non-negative,

$$\begin{aligned}\int_{-\infty}^{\infty} \widehat{f}_k(x)\,dx &\ge \int_{X_{(n)}}^{\infty} \widehat{f}_k(x)\,dx \\ &= \frac{k}{2n}\int_{X_{(n)}}^{\infty} \frac{dx}{x - X_{(n-k+1)}}.\end{aligned}$$

Step 3. The integral is logarithmically divergent:

$$\int_{X_{(n)}}^{T} \frac{dx}{x - X_{(n-k+1)}} = \log\{T - X_{(n-k+1)}\} - \log\{X_{(n)} - X_{(n-k+1)}\}.$$

As $T \to \infty$, the first logarithm tends to infinity. Thus

$$\boxed{\int_{-\infty}^{\infty} \widehat{f}_k(x)\,dx = \infty.}$$

Step 4. To obtain a classifier, estimate the class-conditional densities separately. If Class j has prior π_j and nearest-neighbor density estimate $\widehat{f}_{j,k}(x)$, use

$$\widehat{\delta}(x) = \underset{j}{\operatorname{argmax}}\{\pi_j \widehat{f}_{j,k}(x)\}.$$

If a common neighborhood is used around x, the estimated posterior probability of Class j is the proportion of the k neighbors belonging to that class. Thus the rule chooses the class with the largest number of representatives among the neighbors. For $k = 1$, it simply assigns the label of the closest training observation.

Final conclusion.
The one-dimensional k-NN density estimate has a $1/x$ tail, so its integral diverges. The nearest-neighbor classifier follows from the Bayes rule by replacing the unknown class densities with nearest-neighbor density estimates; with a common neighborhood this reduces to majority vote among the nearest observations.

Question 50

Whole question from the assignment.

> Show that the classifier obtained from a single-layer perceptron model is a linear classifier. Show that minimising the softmax loss function for a single-layer perceptron model is equivalent to maximising the likelihood function in logistic regression.

Theory required for this solution.
A single-layer perceptron forms an affine score from the input:

$$\eta(x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_d x_d.$$

For binary classification, the sigmoid transformation is

$$\sigma(t) = \frac{1}{1 + e^{-t}}, \qquad \widehat{P}(Y = 1 \mid x) = \sigma(\eta(x)).$$

The sigmoid is strictly increasing and equals $1/2$ at $t = 0$. Logistic regression assumes a Bernoulli response with this success probability. Its likelihood is the product of the Bernoulli probabilities.

Step 1. The single-layer perceptron predicts Class 1 when

$$\widehat{P}(Y = 1 \mid x) > \frac{1}{2}.$$

Using the sigmoid expression,

$$\frac{e^{\eta(x)}}{1 + e^{\eta(x)}} > \frac{1}{2} \quad \Longleftrightarrow \quad e^{\eta(x)} > 1 \quad \Longleftrightarrow \quad \eta(x) > 0.$$

The decision boundary is therefore

$$\beta_0 + \beta_1 x_1 + \cdots + \beta_d x_d = 0,$$

which is a hyperplane. Hence the classifier is linear.

Step 2. Let $p_i = \sigma(\eta_i)$, where $\eta_i = \beta_0 + \sum_{j=1}^{d} \beta_j x_{ij}$. Under the Bernoulli logistic model,

$$L(\beta) = \prod_{i=1}^{n} p_i^{y_i} (1 - p_i)^{1-y_i}.$$

Taking logarithms,

$$\ell(\beta) = \sum_{i=1}^{n} \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right].$$

Step 3. The binary softmax/cross-entropy loss is

$$\mathcal{L}_{\text{softmax}}(\beta) = -\sum_{i=1}^{n} \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right].$$

Comparing with the log-likelihood,

$$\mathcal{L}_{\text{softmax}}(\beta) = -\ell(\beta).$$

Therefore minimising this loss is exactly the same optimisation problem as maximising the logistic-regression likelihood.

Final conclusion.

A single-layer perceptron with a sigmoid output has a hyperplane decision boundary. Its binary softmax/cross-entropy loss is the negative logistic-regression log-likelihood:

$$\boxed{\operatorname*{argmin}_{\beta} \mathcal{L}_{\text{softmax}}(\beta) = \operatorname*{argmax}_{\beta} L(\beta).}$$

Question 51

Whole question from the assignment.

> Describe the logistic regression method for a classification problem involving more than two classes. What is the GLMNET classifier?

Theory required for this solution.

For J classes, logistic regression models class probabilities through a collection of linear predictors and a softmax normalisation. A convenient identifiable formulation uses Class J as the reference class:

$$\log \frac{P(Y = j \mid x)}{P(Y = J \mid x)} = \beta_{j0} + \beta_j^\top x, \qquad j = 1, \ldots, J - 1.$$

The corresponding probabilities must be non-negative and sum to one. Ridge and LASSO regularisation add, respectively, squared-ℓ_2 and absolute-ℓ_1 penalties to the likelihood objective.

Step 1. For $j = 1, \ldots, J - 1$, define

$$\eta_j(x) = \beta_{j0} + \beta_j^\top x, \qquad \eta_J(x) = 0.$$

The multinomial logistic probabilities are

$$\widehat{P}(Y = j \mid x) = \frac{e^{\eta_j(x)}}{1 + \sum_{k=1}^{J-1} e^{\eta_k(x)}}, \quad j = 1, \ldots, J - 1,$$

and

$$\widehat{P}(Y = J \mid x) = \frac{1}{1 + \sum_{k=1}^{J-1} e^{\eta_k(x)}}.$$

Step 2. Estimate all coefficient vectors jointly by maximising the multinomial likelihood. If $y_{ij} = 1$ when observation i belongs to Class j and 0 otherwise, the likelihood is

$$L(\beta) = \prod_{i=1}^{n} \prod_{j=1}^{J} \widehat{P}(Y_i = j \mid x_i)^{y_{ij}}.$$

The fitted class is the one with the largest estimated probability:

$$\widehat{\delta}(x) = \underset{1 \le j \le J}{\operatorname{argmax}}\, \widehat{P}(Y = j \mid x).$$

Step 3. Count the free parameters. There are $J - 1$ non-reference class equations, each with an intercept and d slopes. Thus the usual reference-class parameterisation has $(J - 1)(d + 1)$ free coefficients.

Step 4. A GLMNET classifier is a regularised logistic classifier fitted by adding a combination of ridge and LASSO penalties to the negative log-likelihood. A common objective is

$$-\ell(\beta) + \lambda \left\{ \frac{1 - \alpha}{2} \sum_{j} \|\beta_j\|_2^2 + \alpha \sum_{j} \|\beta_j\|_1 \right\},$$

with the intercepts ordinarily left unpenalised. The parameter λ controls the overall penalty strength, while α controls the ridge–LASSO mixture. In the multiclass setting, the likelihood component is multinomial logistic likelihood.

Why this step leads to the next step. The logistic likelihood handles categorical outcomes, while the penalty controls coefficient size and can set some coefficients exactly to zero. This is useful when many predictors are present or when predictors are correlated.

Final conclusion.

Multiclass logistic regression estimates J probabilities through a softmax model and classifies by the largest fitted probability. With a reference class, the number of free parameters is $(J - 1)(d + 1)$. A GLMNET classifier is the corresponding likelihood-based logistic classifier with ridge/LASSO (elastic-net) regularisation.

Question 52

Whole question from the assignment.

> Describe how you will use a multi-layer perceptron model for a classification problem involving more than two classes. Will it lead to nonlinear class boundaries?

Theory required for this solution.

A multilayer perceptron (MLP) has an input layer, one or more hidden layers, and an output layer. Each hidden unit applies a nonlinear transformation to an affine combination of the

preceding layer. The course notes represent a hidden unit as

$$H_\ell = \sigma_\ell\left(\alpha_{0\ell} + \sum_{j=1}^{d} \alpha_{j\ell} X_j\right).$$

For J classes, use J output scores and normalise them using the softmax transformation so that the outputs are estimated class probabilities.

Step 1. Encode the class label using one-hot indicators:

$$Y_{ij} = \begin{cases} 1, & \text{observation } i \text{ belongs to class } j, \\ 0, & \text{otherwise.} \end{cases}$$

There are J output nodes, one per class.

Step 2. For hidden unit ℓ, compute

$$H_{i\ell} = \sigma_\ell\left(\alpha_{0\ell} + \sum_{r=1}^{d} \alpha_{r\ell} X_{ir}\right), \qquad \ell = 1, \ldots, K.$$

The K hidden-unit outputs are then passed to the output layer. For class j, define the score

$$\eta_{ij} = \beta_{0j} + \sum_{\ell=1}^{K} \beta_{\ell j} H_{i\ell}.$$

Step 3. Convert scores to probabilities by softmax:

$$\widehat{P}(Y_i = j \mid x_i) = \frac{e^{\eta_{ij}}}{\sum_{r=1}^{J} e^{\eta_{ir}}}.$$

Train all weights by minimising the multiclass cross-entropy loss

$$-\sum_{i=1}^{n}\sum_{j=1}^{J} Y_{ij} \log \widehat{P}(Y_i = j \mid x_i).$$

A gradient-based method can be used because the hidden and output transformations make the objective nonlinear in the network weights.

Step 4. The model generally produces nonlinear class boundaries. Each hidden unit transforms the original predictors nonlinearly, and the output scores are combinations of these transformed features. A boundary between classes j and k is determined by $\eta_j(x) = \eta_k(x)$, which is generally nonlinear in the original x.

Final conclusion.

Use one-hot class outputs, one or more hidden layers with nonlinear activation functions, a J-node softmax output layer, and cross-entropy training. The nonlinear hidden transformations allow nonlinear class boundaries. If all activations were linear, stacking layers would still yield only an affine model and would not create nonlinear boundaries.

Question 53

Whole question from the assignment.

> A data set contains the information on the age (X_1), gender (X_2), typing speed (X_3), concentration level (X_4), daily working hours (X_5) and average number of mistakes (Y) on a page for different typists. Use an appropriate statistical model for this data set for predicting the average number of mistakes based on other covariates.

Theory required for this solution.

The supplied GLM notes describe Poisson regression for non-negative count responses. Under a Poisson model,

$$Y_i \mid x_i \sim \text{Poisson}(\mu_i), \qquad \log \mu_i = \eta_i = \beta_0 + \sum_{j=1}^{5} \beta_j x_{ij}.$$

The log link ensures that the fitted mean is positive. A categorical predictor such as gender is represented by an indicator variable, with one category taken as the reference.

Step 1. Identify the response. The number of mistakes on a page is a non-negative count. Under the natural count-data interpretation, use a Poisson regression model. (If the recorded response is a fractional average over many pages rather than a count or count rate, the response distribution should be checked before adopting Poisson; the assignment provides no raw data for that check.)

Step 2. Encode gender. For example, set $X_2 = 1$ for one gender category and $X_2 = 0$ for the reference category. The other predictors remain numeric measurements.

Step 3. Specify the conditional distribution and link:

$$Y_i \mid X_{1i}, \ldots, X_{5i} \sim \text{Poisson}(\mu_i),$$

$$\log \mu_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \beta_5 X_{5i}.$$

Equivalently,

$$\mu_i = \exp\left(\beta_0 + \sum_{j=1}^{5} \beta_j X_{ji}\right).$$

Step 4. Estimate $\beta_0, \ldots, \beta_5$ by maximising the Poisson likelihood. For a new typist with covariate vector x_0, the predicted mean number of mistakes per page is

$$\widehat{\mu}_0 = \exp\left(\widehat{\beta}_0 + \widehat{\beta}_1 x_{01} + \cdots + \widehat{\beta}_5 x_{05}\right).$$

Step 5. Interpret the coefficients. Holding the other predictors fixed, a one-unit increase in X_j multiplies the expected count by e^{β_j}. For the gender indicator, e^{β_2} is the ratio of expected mistake counts between the coded category and the reference category, all else equal.

Final conclusion.

The appropriate course-level model for a count response is Poisson regression with a log link:

$$\boxed{Y_i \mid x_i \sim \text{Poisson}(\mu_i), \qquad \log \mu_i = \beta_0 + \sum_{j=1}^{5} \beta_j X_{ji}.}$$

Gender must be represented by a suitable indicator variable. The fitted conditional mean gives the prediction for the average number of mistakes.

Question 54

Whole question from the assignment.

Consider a regression problem, where the response variable Y takes only two values, 0 and 1, and they are arranged below according to increasing values of a covariate X_1.

X_1	0.6	1.4	2.2	2.5	2.9	3.2	3.8	4.7	5.3	5.6	6.5	7.0	8.6	9.4	10.1
Y	0	0	0	0	0	1	0	1	0	1	0	1	1	1	1

Find the best split based on X_1 using the Gini index as the impurity function. Does your answer change if the entropy or the misclassification rate is used instead?

Theory required for this solution.
For a node containing class proportions p and $1-p$, the three impurity functions are

$$G(p) = 2p(1-p), \qquad H(p) = -p\log_2 p - (1-p)\log_2(1-p),$$

$$M(p) = 1 - \max(p, 1-p) = \min(p, 1-p),$$

with $0\log 0$ interpreted as 0. A candidate split at threshold t divides the ordered observations into a left node and a right node. The weighted post-split impurity is

$$I_{\text{split}}(t) = \frac{n_L}{n} I_L + \frac{n_R}{n} I_R.$$

The best split minimises this weighted impurity.

Step 1. Since X_1 is ordered, it is sufficient to test thresholds between consecutive observed values. The candidate thresholds are the midpoints

$$1.00, 1.80, 2.35, 2.70, 3.05, 3.50, 4.25, 5.00, 5.45, 6.05, 6.75, 7.80, 9.00, 9.75.$$

Step 2. At $t = 3.05$, the left node contains the first five observations. All five responses are 0, so $p_L = 0$ and all three impurities are zero in the left node. The right node contains ten observations, of which seven have response 1 and three have response 0. Thus $p_R = 7/10$.

Step 3. Calculate the weighted Gini index:

$$\begin{aligned} I_G(3.05) &= \frac{5}{15}\,2(0)(1) + \frac{10}{15}\,2\left(\frac{7}{10}\right)\left(\frac{3}{10}\right) \\ &= \frac{2}{3}(0.42) = 0.28. \end{aligned}$$

For comparison, the corresponding entropy is

$$I_H(3.05) = \frac{10}{15}\left[-\frac{7}{10}\log_2\frac{7}{10} - \frac{3}{10}\log_2\frac{3}{10}\right] \approx 0.58753,$$

and the weighted misclassification impurity is

$$I_M(3.05) = \frac{10}{15}\min\left(\frac{7}{10}, \frac{3}{10}\right) = 0.20.$$

Step 4. Compare all candidate splits. The table reports the left and right class-1 counts and the weighted post-split impurities (rounded to six decimal places).

Threshold t	Class 1 left	Class 1 right	Gini	Entropy	Misclass.
1.00	0/1	7/14	0.466667	0.933333	0.466667
1.80	0/2	7/13	0.430769	0.862964	0.400000
2.35	0/3	7/12	0.388889	0.783895	0.333333
2.70	0/4	7/11	0.339394	0.693484	0.266667
3.05	0/5	7/10	**0.280000**	**0.587527**	0.200000
3.50	1/6	6/9	0.377778	0.810986	0.266667
4.25	1/7	6/8	0.314286	0.708796	0.200000
5.00	2/8	5/7	0.390476	0.835471	0.266667
5.45	2/9	5/6	0.318519	0.718532	0.200000
6.05	3/10	4/5	0.386667	0.828170	0.266667
6.75	3/11	4/4	0.290909	0.619924	0.200000
7.80	4/12	3/3	0.355556	0.734637	0.266667
9.00	5/13	2/2	0.410256	0.833072	0.333333
9.75	6/14	1/1	0.457143	0.919546	0.400000

Step 5. The Gini and entropy columns have their unique minimum at $t = 3.05$. The misclassification column has minimum 0.20 at four thresholds: 3.05, 4.25, 5.45, and 6.75. Thus the split selected by the misclassification impurity is not unique.

Final conclusion.

Using Gini, the best split is

$$\boxed{X_1 \leq 3.05 \quad \text{versus} \quad X_1 > 3.05.}$$

Entropy selects the same unique split. Misclassification impurity attains the same minimum value at four thresholds, so it does not determine a unique split, although 3.05 is one of its minimisers.

Question 55

Whole question from the assignment.

Let $\widehat{f}_{k,n}$ be the kth nearest-neighbour estimate of a regression function based on n ($n > 10$) observations with distinct x-values. If $\widehat{f}_{1,n}(x) \leq 1$ for all $x \in \mathbb{R}^d$, show that $\widehat{f}_{2,n}(x) \leq 1$ for all $x \in \mathbb{R}^d$. Is the converse true? Justify your answer.

Theory required for this solution.

The k-nearest-neighbour regression estimate is the average of the response values attached to the k closest training observations:

$$\widehat{f}_{k,n}(x) = \frac{1}{k} \sum_{i \in N_k(x)} Y_i,$$

where $N_k(x)$ is the set of indices of the k nearest training predictor vectors to x. Distinct predictor vectors ensure that, at a training location x_i, the unique nearest neighbour is observation i itself.

Step 1. Assume $\widehat{f}_{1,n}(x) \le 1$ for every x. Evaluate this inequality at each observed predictor vector x_i. Since x_i is its own unique nearest neighbour,

$$\widehat{f}_{1,n}(x_i) = Y_i.$$

Therefore

$$Y_i \le 1 \qquad \text{for every } i = 1, \dots, n.$$

Step 2. At any query point x, the 2-NN estimate is the average of two responses. Each response is at most 1 by Step 1. Hence

$$\widehat{f}_{2,n}(x) = \frac{Y_{(1)}(x) + Y_{(2)}(x)}{2} \le \frac{1+1}{2} = 1.$$

This proves the forward implication.

Step 3. The converse is false. Construct a data set with distinct predictor vectors, one response equal to 2, and all remaining responses equal to 0. For example, take any $n > 10$ distinct points $x_1, \dots, x_n$ and set

$$Y_1 = 2, \qquad Y_2 = \cdots = Y_n = 0.$$

Every pair of nearest-neighbour responses contains at most one value equal to 2 and the other value is 0. Thus every 2-NN estimate is either 0 or 1, so it is at most 1 for all x. However, at $x = x_1$,

$$\widehat{f}_{1,n}(x_1) = Y_1 = 2 > 1.$$

Final conclusion.

$$\boxed{\widehat{f}_{1,n}(x) \le 1 \ \forall x \quad \Longrightarrow \quad \widehat{f}_{2,n}(x) \le 1 \ \forall x,}$$

but the converse is false. A 2-NN average can hide a single response larger than 1 by averaging it with a sufficiently small second response.

Question 56

Whole question from the assignment.

> Consider the classification problem given in Question no. 30. Give an ideal perceptron model (with appropriate network architecture and transformation function) for this problem.

Theory required for this solution.

In Question 30, Class 1 is an equal mixture with component means $(1,1)$ and $(-1,-1)$, while Class 2 is an equal mixture with component means $(-1,1)$ and $(1,-1)$. Each component has covariance $(1/4)I_2$. The Bayes comparison reduces to the sign of x_1x_2: Class 1 corresponds to the same-sign quadrants and Class 2 to the opposite-sign quadrants. A single affine perceptron cannot represent this pattern because the two same-sign quadrants are not one half-plane. A multilayer perceptron with threshold activations can represent it.

Step 1. Identify the required decision pattern. The ideal classifier is

$$\delta(x) = \begin{cases} 1, & x_1x_2 > 0, \\ 2, & x_1x_2 < 0. \end{cases}$$

Points on the axes lie on the decision boundary and may be assigned to either class because they have probability zero under the continuous distributions.

Step 2. Use the threshold transformation

$$H(t) = \begin{cases} 1, & t > 0, \\ 0, & t \leq 0. \end{cases}$$

The first hidden layer has two units:

$$h_1 = H(x_1), \qquad h_2 = H(x_2).$$

These units encode whether each coordinate is positive.

Step 3. Use a second hidden layer to detect the two same-sign cases. Define

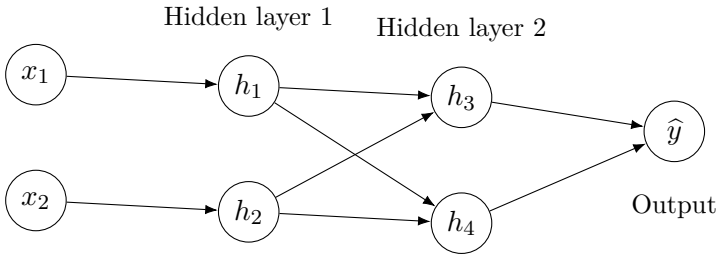
$$h_3 = H(h_1 + h_2 - 3/2), \qquad h_4 = H(1/2 - h_1 - h_2).$$

Then $h_3 = 1$ exactly when $(h_1, h_2) = (1, 1)$, and $h_4 = 1$ exactly when $(h_1, h_2) = (0, 0)$.

Step 4. The output unit takes the logical OR of these two detectors:

$$\widehat{y} = H(h_3 + h_4 - 1/2).$$

If $\widehat{y} = 1$, predict Class 1; otherwise predict Class 2. This implements the same-sign versus opposite-sign quadrant rule exactly.



Final conclusion.
An ideal exact-threshold network has two input coordinates, two threshold units in the first hidden layer, two quadrant-detector units in the second hidden layer, and one OR output unit. Its output is Class 1 when x_1 and x_2 have the same sign and Class 2 when they have opposite signs. A single-layer perceptron is insufficient because the Bayes decision region is not a single half-plane.

Question 57

Whole question from the assignment.

> Consider the same problem (given in Question no. 30). Suppose that we have some observations from each of the two classes. Now, a person assumes the underlying distributions for both classes to be normal and carries out quadratic discriminant analysis. How will the classifier perform on this data set? Justify your answer.

Theory required for this solution.
QDA models each class by a multivariate normal distribution with its own mean vector and covariance matrix. Its decision rule compares the two fitted normal densities. QDA uses class-specific covariance matrices, so its log-density comparison generally contains quadratic terms. The class distribution in Question 30 is actually a mixture of two normal components, not one normal distribution per class. Nevertheless, the first two moments can still produce a QDA boundary that agrees with the true Bayes boundary.

Step 1. Compute the population mean and covariance of Class 1. Its component means are $(1,1)^\top$ and $(-1,-1)^\top$, each with covariance $\frac{1}{4}I_2$. The overall mean is

$$\mu_1 = \frac{1}{2}\begin{pmatrix}1\\1\end{pmatrix} + \frac{1}{2}\begin{pmatrix}-1\\-1\end{pmatrix} = 0.$$

The covariance is the within-component covariance plus the covariance of the component means:

$$\Sigma_1 = \frac{1}{4}I_2 + \begin{pmatrix}1 & 1\\1 & 1\end{pmatrix} = \begin{pmatrix}5/4 & 1\\1 & 5/4\end{pmatrix}.$$

Step 2. Do the same for Class 2. Its component means are $(-1,1)^\top$ and $(1,-1)^\top$, so its mean is also zero. Its covariance is

$$\Sigma_2 = \frac{1}{4}I_2 + \begin{pmatrix}1 & -1\\-1 & 1\end{pmatrix} = \begin{pmatrix}5/4 & -1\\-1 & 5/4\end{pmatrix}.$$

The determinants are equal:

$$|\Sigma_1| = |\Sigma_2| = \frac{25}{16} - 1 = \frac{9}{16}.$$

Step 3. Under the population QDA fit, the means, determinants, and equal priors cancel from the log-density comparison except for the quadratic forms. The inverse matrices are

$$\Sigma_1^{-1} = \frac{16}{9}\begin{pmatrix}5/4 & -1\\-1 & 5/4\end{pmatrix}, \qquad \Sigma_2^{-1} = \frac{16}{9}\begin{pmatrix}5/4 & 1\\1 & 5/4\end{pmatrix}.$$

QDA assigns Class 1 when

$$-\frac{1}{2}x^\top\Sigma_1^{-1}x > -\frac{1}{2}x^\top\Sigma_2^{-1}x,$$

which is equivalent to

$$x^\top(\Sigma_2^{-1} - \Sigma_1^{-1})x > 0.$$

Now

$$\Sigma_2^{-1} - \Sigma_1^{-1} = \frac{32}{9}\begin{pmatrix}0 & 1\\1 & 0\end{pmatrix},$$

so the inequality becomes

$$\frac{64}{9}x_1x_2 > 0 \quad\Longleftrightarrow\quad x_1x_2 > 0.$$

Step 4. The true Bayes classifier for the mixture distributions in Question 30 also assigns Class 1 when $x_1x_2 > 0$ and Class 2 when $x_1x_2 < 0$. Hence, at the population-moment level, QDA gives the same decision boundary as the Bayes rule, even though the normality assumption within each class is not literally correct.

Final conclusion.
QDA is a fortuitous match in this example: the class distributions are mixtures rather than single normal distributions, but their means and covariance matrices yield the exact same quadratic decision boundary $x_1 x_2 = 0$ as the Bayes classifier. With finite samples, estimated means and covariance matrices fluctuate, so the fitted QDA boundary will only approximate this population boundary; exact finite-sample agreement is not guaranteed.

Question 58

Whole question from the assignment.

> Describe how the Nadaraya–Watson estimator of a regression function can be meaningfully used for a binary classification problem. How can it be used for classification problems involving more than two classes?

Theory required for this solution.
The Nadaraya–Watson estimator is a locally weighted average:

$$\widehat{m}(x) = \frac{\sum_{i=1}^{n} K\left(\frac{x-X_i}{h}\right) Y_i}{\sum_{i=1}^{n} K\left(\frac{x-X_i}{h}\right)}.$$

For a binary response $Y \in \{0, 1\}$,

$$\mathbb{E}(Y \mid X = x) = \mathbb{P}(Y = 1 \mid X = x).$$

Thus a regression estimate of the conditional mean is also an estimate of the posterior class probability.

Step 1. For binary classification, code the classes as 0 and 1. Apply the Nadaraya–Watson estimator:

$$\widehat{p}_1(x) = \frac{\sum_{i=1}^{n} K((x - X_i)/h) Y_i}{\sum_{i=1}^{n} K((x - X_i)/h)}.$$

Because Y_i is an indicator of Class 1, the numerator is the total kernel weight contributed by Class 1 observations.

Step 2. Estimate the other posterior probability by

$$\widehat{p}_0(x) = 1 - \widehat{p}_1(x).$$

Choose Class 1 if $\widehat{p}_1(x) > 1/2$, and Class 0 otherwise. Equivalently, choose the class with the larger estimated posterior probability.

Step 3. For J classes, use one-hot indicators $Y_{ij} = I(Y_i = j)$. For each class j, estimate

$$\widehat{p}_j(x) = \frac{\sum_{i=1}^{n} K((x - X_i)/h) I(Y_i = j)}{\sum_{i=1}^{n} K((x - X_i)/h)}.$$

These estimates are non-negative and sum to one, because every observation belongs to exactly one class.

Step 4. Use the multiclass decision rule

$$\widehat{\delta}(x) = \underset{1\le j\le J}{\operatorname{argmax}}\, \widehat{p}_j(x).$$

In words, assign x to the class whose one-hot response has the largest locally weighted average.

Final conclusion.
Nadaraya–Watson regression becomes classification when the response is replaced by class-indicator variables. For two classes, threshold the estimated Class 1 probability at $1/2$; for J classes, estimate all J posterior probabilities and select the largest.

Question 59

Whole question from the assignment.

The central region of the d-dimensional standard normal distribution with density ϕ_d is defined as

$$C_d = \left\{ x : \frac{\phi_d(x)}{\phi_d(0)} > 0.01 \right\}.$$

Show that if X_d denotes a d-dimensional standard normal variate, $\mathbb{P}(X_d \in C_d) \to 0$ as d grows to infinity.

Theory required for this solution.
The standard d-dimensional normal density is

$$\phi_d(x) = (2\pi)^{-d/2} \exp(-\|x\|^2/2).$$

Therefore its ratio to the density at the origin depends only on the squared norm. For $X_d \sim N_d(0, I_d)$,

$$\|X_d\|^2 \sim \chi^2_d, \qquad \mathbb{E}\|X_d\|^2 = d, \qquad \operatorname{Var}(\|X_d\|^2) = 2d.$$

Step 1. Simplify the definition of the central region:

$$\frac{\phi_d(x)}{\phi_d(0)} = e^{-\|x\|^2/2} > 0.01.$$

Taking logarithms,

$$-\frac{1}{2}\|x\|^2 > \log(0.01) = -\log 100,$$

so

$$C_d = \{x : \|x\|^2 < 2\log 100\}.$$

Let $c = 2\log 100$, a fixed positive constant independent of d.

Step 2. Since $\|X_d\|^2 \sim \chi^2_d$,

$$\mathbb{P}(X_d \in C_d) = \mathbb{P}(\chi^2_d < c).$$

The chi-square random variable has mean d and variance $2d$.

Step 3. For $d > c$, the event $\{\chi^2_d < c\}$ implies $|\chi^2_d - d| > d - c$. By Chebyshev's inequality,

$$\mathbb{P}(\chi^2_d < c) \le \frac{2d}{(d-c)^2} \longrightarrow 0.$$

Final conclusion.

$$\boxed{\mathbb{P}(X_d \in C_d) \longrightarrow 0.}$$

The density remains maximal at the origin, but the probability mass of any fixed-radius central ball vanishes as dimension increases.

Question 60

Whole question from the assignment.

> Give an example of a binary classification problem with equal priors, where the usual nearest-neighbor classifier classifies almost all future observations to a single class. Justify your answer.

Theory required for this solution.
The 1-nearest-neighbour classifier assigns a new point the label of the closest training point. Therefore the classifier's predictions depend on the geometry of the labelled training locations, not just on the population class priors. A training observation owns the part of the feature space for which it is the nearest training point (its Voronoi cell). If the only Class 1 observation has a very small Voronoi cell, almost every future query can be assigned to Class 2 even when the population priors are equal.

Step 1. Construct a population problem with equal priors and identical class-conditional distributions:

$$\mathbb{P}(C_1) = \mathbb{P}(C_2) = \frac{1}{2}, \qquad X \mid C_1 \sim U([0,1]^2), \quad X \mid C_2 \sim U([0,1]^2).$$

Thus neither class is intrinsically more common in the population, and the feature distribution contains no information about the class label.

Step 2. Consider a possible labelled training sample with one Class 1 observation at $c = (0.5, 0.5)$ and 99 Class 2 observations. Among the 99 Class 2 observations, include the four points

$$c + (0.2, 0), \quad c - (0.2, 0), \quad c + (0, 0.2), \quad c - (0, 0.2),$$

and place the remaining 95 Class 2 observations at any other distinct locations in the unit square. The exact coordinates are only a convenient schematic configuration. If each point is allowed to vary in a sufficiently small neighbourhood of its displayed location, the Voronoi-cell bound below remains below 5%; such neighbourhoods have positive probability under the continuous uniform distributions. The class sample counts may also be fixed by the study design.

Step 3. Determine a region containing the Voronoi cell of the lone Class 1 point. For a query $x = (x_1, x_2)$ to be closer to c than to $c + (0.2, 0)$, it must satisfy

$$\|x - c\|^2 \le \|x - c - (0.2, 0)\|^2 \quad \Longrightarrow \quad x_1 \le 0.6.$$

Comparison with $c - (0.2, 0)$ similarly gives $x_1 \ge 0.4$. The two vertical neighbours therefore force $0.4 \le x_1 \le 0.6$. The two horizontal neighbours force $0.4 \le x_2 \le 0.6$. Hence the Class 1 Voronoi cell is contained in the square $[0.4, 0.6]^2$, whose area is

$$(0.2)^2 = 0.04.$$

The other Class 2 points can only make this cell smaller.

Step 4. A future observation is uniform on the unit square, regardless of its class. Therefore the probability that it falls in the Class 1 Voronoi cell is at most 0.04. At least 0.96 of future observations are consequently classified as Class 2 by 1-NN. This is the requested phenomenon: the classifier assigns almost all future observations to one class, despite equal population priors.

Final conclusion.

The example has equal priors and identical class-conditional feature distributions, but a realised training set with one Class 1 point whose Voronoi cell occupies at most 4% of the feature-space support, and 99 Class 2 points. Thus at least 96% of future observations are classified as Class 2. The effect is caused by the spatial arrangement and imbalance of the labelled training sample, not by unequal population priors.

Question 61

Whole question from the assignment.

> Consider a classification problem involving observations from two 100-dimensional absolutely continuous probability distributions. If there are 40 observations from the first class and 50 observations from the second class, show that with probability one, there will be a linear classifier which will perfectly separate the observations from the two classes.

Theory required for this solution.

An affine hyperplane in $\mathbb{R}^{100}$ has the form $w^\top x + b = 0$. A finite collection of points drawn from absolutely continuous distributions is in general affine position with probability one, provided the observations are independent. In $\mathbb{R}^d$, any $m \le d+1$ points in general affine position are affinely independent. Here there are $m = 40 + 50 = 90$ points and $d + 1 = 101$ augmented coordinates when the intercept is included.

Step 1. Combine all 90 observations and denote them by $x_1, \dots, x_{90} \in \mathbb{R}^{100}$. Attach labels $y_i = +1$ to the 40 observations from Class 1 and $y_i = -1$ to the 50 observations from Class 2.

Step 2. Form augmented vectors

$$z_i = \begin{pmatrix} x_i \\ 1 \end{pmatrix} \in \mathbb{R}^{101}.$$

Because the original distributions are absolutely continuous, the event that a newly drawn observation falls in the affine span of fewer than 101 previous generic observations has probability zero. Repeating this argument shows that the 90 augmented vectors are linearly independent with probability one.

Step 3. Put the augmented vectors as rows of a 90×101 matrix Z. With probability one, Z has full row rank 90. Hence the linear system

$$Z\theta = y, \qquad \theta = \begin{pmatrix} w \\ b \end{pmatrix},$$

has a solution because a full-row-rank matrix maps onto $\mathbb{R}^{90}$.

Step 4. For this solution,

$$w^\top x_i + b = y_i \in \{-1, +1\}.$$

Thus

$$y_i(w^\top x_i + b) = 1 > 0 \qquad \text{for all } i,$$

which is strict linear separation of all the training observations.

Final conclusion.

Since $90 < 101$, the 90 augmented observations are linearly independent with probability one. There is therefore an affine hyperplane separating the two labelled groups perfectly with probability one.

Question 62

Whole question from the assignment.

> Consider the setup considered in Question no. 61. Show that in this case, the maximum likelihood estimates of the parameters of the logistic regression model does not exist.

Theory required for this solution.

For logistic regression with labels $y_i \in \{-1, +1\}$ and parameter vector $\theta = (w^\top, b)^\top$, write

$$P(Y_i = y_i \mid x_i; \theta) = \frac{1}{1 + e^{-y_i(w^\top x_i + b)}}.$$

If the data are strictly linearly separable, there is a vector θ_0 such that every signed margin $y_i \theta_0^\top z_i$ is positive. Scaling a separating vector increases all margins simultaneously.

Step 1. By Question 61, with probability one there exists θ_0 satisfying

$$y_i \theta_0^\top z_i > 0 \qquad \text{for every } i.$$

The logistic log-likelihood is

$$\ell(\theta) = \sum_{i=1}^{n} \log\left\{\frac{1}{1 + e^{-y_i \theta^\top z_i}}\right\} = -\sum_{i=1}^{n} \log\left(1 + e^{-y_i \theta^\top z_i}\right).$$

Step 2. Consider the one-parameter sequence $\theta_t = t\theta_0$ with $t > 0$. Then

$$\ell(t\theta_0) = -\sum_{i=1}^{n} \log\left(1 + e^{-t y_i \theta_0^\top z_i}\right).$$

Every margin $y_i \theta_0^\top z_i$ is positive, so as $t \to \infty$,

$$e^{-t y_i \theta_0^\top z_i} \longrightarrow 0 \quad \Longrightarrow \quad \log\left(1 + e^{-t y_i \theta_0^\top z_i}\right) \longrightarrow 0.$$

Consequently,

$$\ell(t\theta_0) \longrightarrow 0.$$

Step 3. For every finite t, each term in the log-likelihood is strictly negative, so $\ell(t\theta_0) < 0$. The supremum 0 is approached as $t \to \infty$ but is not attained by any finite parameter vector. Hence no finite maximum-likelihood estimate exists.

Final conclusion.
Under complete linear separation, the logistic likelihood can always be increased by multiplying a separating coefficient vector by a larger positive constant. The supremum is approached only as the coefficient norm diverges. Therefore the finite logistic-regression MLE does not exist.

Question 63

Whole question from the assignment.

> Consider a classification problem involving J classes. If the observations are d-dimensional, how many parameters are there in the logistic regression model? Suppose that you know how to fit a generalized linear model with logit link function. How will you modify your method to fit a generalized additive model with the logit link?

Theory required for this solution.
For J classes, choose one reference class. Each of the other $J-1$ classes has a log-odds equation relative to the reference class. With d predictors, each equation has one intercept and d slopes. A generalized additive model replaces the linear contribution of a predictor by a smooth univariate function of that predictor:

$$\eta(x) = \beta_0 + f_1(x_1) + \cdots + f_d(x_d).$$

For a binary response, the logit link is $g(p) = \log\{p/(1-p)\}$.

Step 1. Count the multiclass logistic parameters. For $j = 1, \ldots, J-1$,

$$\log \frac{P(Y = j \mid x)}{P(Y = J \mid x)} = \beta_{j0} + \sum_{r=1}^{d} \beta_{jr} x_r.$$

There are $d+1$ parameters for each of $J-1$ equations. The total is

$$\boxed{(J-1)(d+1)}.$$

Equivalently, one could write J score vectors and impose an identifiability constraint; the number of free parameters remains the same.

Step 2. For a binary GAM with logit link, replace the linear predictor by

$$\eta(x) = \beta_0 + f_1(x_1) + \cdots + f_d(x_d),$$

so that

$$\log \frac{P(Y = 1 \mid x)}{1 - P(Y = 1 \mid x)} = \beta_0 + \sum_{r=1}^{d} f_r(x_r).$$

The functions f_r describe potentially nonlinear effects of individual predictors while keeping the model additive across predictors.

Step 3. To fit the GAM using a GLM fitting method, represent each smooth function by a finite basis expansion, for example

$$f_r(x_r) = \sum_{k=1}^{K_r} \gamma_{rk} B_{rk}(x_r),$$

where B_{rk} are chosen basis functions. The predictor becomes linear in the unknown basis coefficients:

$$\eta(x) = \beta_0 + \sum_{r=1}^{d} \sum_{k=1}^{K_r} \gamma_{rk} B_{rk}(x_r).$$

Construct an expanded design matrix containing these basis columns and fit the resulting Bernoulli GLM with logit link. If smoothness penalties are used, include them in the optimisation and choose their strengths appropriately.

Step 4. For multiclass additive logistic regression, use a separate additive predictor for each non-reference class:

$$\log \frac{P(Y = j \mid x)}{P(Y = J \mid x)} = \beta_{j0} + \sum_{r=1}^{d} f_{jr}(x_r), \qquad j = 1, \ldots, J-1.$$

Fit the expanded predictors jointly through the multinomial likelihood.

Final conclusion.
The reference-class multiclass logistic model has $(J-1)(d+1)$ free parameters. To obtain a GAM with logit link, replace each linear predictor by an additive sum of smooth functions, represent those functions using basis expansions, and apply the GLM fitting machinery to the resulting expanded design.

Question 64

Whole question from the assignment.

> What is probit regression? Describe the iteratively reweighted least squares method for fitting a probit regression model for a binary classification problem.

Theory required for this solution.
For a binary response $Y \in \{0, 1\}$, probit regression uses the standard normal cdf as the response transformation:

$$\mathbb{P}(Y = 1 \mid x) = \Phi(\eta), \qquad \eta = x^\top \beta,$$

so the link is $g(p) = \Phi^{-1}(p)$. Write $\phi(\eta)$ for the standard normal density and $\mu = \Phi(\eta)$. The Bernoulli variance is $\mu(1-\mu)$. The GLM score and Fisher-scoring equations give an iteratively reweighted least-squares update.

Step 1. State the model. For observations $i = 1, \ldots, n$,

$$Y_i \mid x_i \sim \text{Bernoulli}(\mu_i), \qquad \mu_i = \Phi(\eta_i), \qquad \eta_i = x_i^\top \beta.$$

The fitted probability is between 0 and 1 because it is a normal cdf.

Step 2. For a general GLM with link $\eta = g(\mu)$, define the working response and weight as

$$z_i = \eta_i + (y_i - \mu_i) g'(\mu_i), \qquad w_i = \frac{1}{\mu_i(1-\mu_i)\{g'(\mu_i)\}^2}.$$

These are obtained by a first-order expansion of the link around the current fitted mean, together with the Bernoulli variance.

Step 3. For the probit link $g(\mu) = \Phi^{-1}(\mu)$, the inverse-link derivative is

$$\frac{d\mu}{d\eta} = \phi(\eta), \qquad g'(\mu) = \frac{d\eta}{d\mu} = \frac{1}{\phi(\eta)}.$$

Substitute this derivative into the working response and weight:

$$\boxed{z_i = \eta_i + \frac{y_i - \mu_i}{\phi(\eta_i)},}$$

$$\boxed{w_i = \frac{\phi(\eta_i)^2}{\mu_i(1-\mu_i)}.}$$

Step 4. The IRLS algorithm is now explicit.

1. Start with an initial coefficient vector $\beta^{(0)}$.
2. At iteration t, compute $\eta_i^{(t)} = x_i^\top \beta^{(t)}$ and $\mu_i^{(t)} = \Phi(\eta_i^{(t)})$.
3. Compute $z_i^{(t)}$ and $w_i^{(t)}$ from the two boxed formulas.
4. Let X be the design matrix and $W^{(t)} = \operatorname{diag}(w_1^{(t)}, \dots, w_n^{(t)})$. Solve the weighted least-squares problem:
$$\beta^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} z^{(t)}.$$
5. Repeat until the coefficients or log-likelihood change by less than a chosen tolerance.

Why this step leads to the next step. At every iteration, the current nonlinear GLM problem is approximated by a weighted linear least-squares problem. Updating the working response and weights improves this local approximation, which is why the procedure is called iteratively reweighted least squares.

Final conclusion.
Probit regression models $P(Y = 1 \mid x)$ as $\Phi(x^\top \beta)$. Its IRLS update uses

$$z_i = \eta_i + \frac{y_i - \mu_i}{\phi(\eta_i)}, \qquad w_i = \frac{\phi(\eta_i)^2}{\mu_i(1-\mu_i)},$$

followed by weighted least squares for the next coefficient vector.

Question 65

Whole question from the assignment.

For each of the following models, determine whether it is a Generalized Additive Model (GAM). Justify your answers by identifying an appropriate link function, where applicable.

1. $\mathbb{E}(Y \mid X_1, X_2) = e^{X_1 + 2X_2}$.
2. $\mathbb{E}(Y \mid X_1, X_2) = e^{X_1 X_2}$.
3. $\log \mathbb{E}(Y \mid X_1, X_2) = \sin(X_1) + X_2^2$.
4. $\log \dfrac{P(Y = 1 \mid X_1, X_2)}{1 - P(Y = 1 \mid X_1, X_2)} = X_1^2 + \log(1 + X_2^2)$.

5. $P(Y = 1 \mid X_1, X_2) = \dfrac{1}{1 + e^{-(X_1 + X_2^2)}}$.
6. $\log \mathbb{E}(Y \mid X_1, X_2) = X_1^2 + X_2^2 + X_1 X_2$.

Assume that the conditional distribution of Y belongs to an appropriate exponential-family distribution, and that the displayed link functions are valid for the response under consideration.

Theory required for this solution.
A generalized additive model has a link-transformed conditional mean of the form

$$g\{\mathbb{E}(Y \mid X_1, X_2)\} = \beta_0 + f_1(X_1) + f_2(X_2),$$

where each f_j depends on one predictor only. For a binary response, the conditional mean equals $P(Y = 1 \mid X_1, X_2)$ and the logit link is $g(p) = \log\{p/(1-p)\}$. For a positive mean, the log link is commonly used. An interaction term that genuinely depends jointly on both variables is not additive in the original predictors.

Step 1. Model 1:

$$\mathbb{E}(Y \mid X_1, X_2) = e^{X_1 + 2X_2}.$$

Use the log link:

$$\log \mathbb{E}(Y \mid X_1, X_2) = X_1 + 2X_2.$$

This is additive with $f_1(X_1) = X_1$ and $f_2(X_2) = 2X_2$. Therefore it **is a GAM**.

Step 2. Model 2:

$$\mathbb{E}(Y \mid X_1, X_2) = e^{X_1 X_2}.$$

Using the log link gives

$$\log \mathbb{E}(Y \mid X_1, X_2) = X_1 X_2.$$

The product is an interaction: its change with X_1 depends on X_2 and vice versa. It cannot be written as a sum of a function of X_1 and a function of X_2 over unrestricted predictor values. Therefore it is **not a GAM** in the stated predictors.

Step 3. Model 3:

$$\log \mathbb{E}(Y \mid X_1, X_2) = \sin(X_1) + X_2^2.$$

This already has the additive form with $f_1(X_1) = \sin(X_1)$ and $f_2(X_2) = X_2^2$. Therefore it **is a GAM**.

Step 4. Model 4:

$$\log \frac{P(Y = 1 \mid X_1, X_2)}{1 - P(Y = 1 \mid X_1, X_2)} = X_1^2 + \log(1 + X_2^2).$$

The displayed link is logit, and the right-hand side is additive. Take

$$f_1(X_1) = X_1^2, \qquad f_2(X_2) = \log(1 + X_2^2).$$

Therefore it **is a GAM**.

Step 5. Model 5:

$$P(Y = 1 \mid X_1, X_2) = \frac{1}{1 + e^{-(X_1 + X_2^2)}}.$$

This is the sigmoid of $X_1 + X_2^2$. Taking the logit gives

$$\log \frac{P(Y = 1 \mid X_1, X_2)}{1 - P(Y = 1 \mid X_1, X_2)} = X_1 + X_2^2.$$

This is additive, with $f_1(X_1) = X_1$ and $f_2(X_2) = X_2^2$. Therefore it **is a GAM**.

Step 6. Model 6:

$$\log \mathbb{E}(Y \mid X_1, X_2) = X_1^2 + X_2^2 + X_1 X_2.$$

The first two terms are additive, but X_1X_2 is an interaction term involving both predictors. It prevents the displayed linear predictor from being a sum of separate univariate functions. Therefore it is **not a GAM** in the stated form.

Final conclusion.

Model	Link used	GAM?
1	log	Yes
2	log	No – interaction X_1X_2
3	log	Yes
4	logit	Yes
5	logit	Yes
6	log	No – interaction X_1X_2

Question 66

Whole question from the assignment.

> Formulate the optimization problem for constructing the linear SVM classifier when the observations from the two classes are linearly separable. With the help of the Lagrangian, construct the corresponding dual problem. If the solution of the dual problem is known, how will you classify a future observation? How will you modify these steps if the observations from the two classes are not linearly separable?

Theory required for this solution.

For labels $y_i \in \{-1, +1\}$, a linear decision boundary is $w^\top x + b = 0$. When the data are linearly separable, scale (w, b) so that the closest observations satisfy the margin constraints

$$y_i(w^\top x_i + b) \geq 1.$$

The geometric margin is inversely proportional to $\|w\|$, so maximising the margin is equivalent to minimising $\frac{1}{2}\|w\|^2$. In the non-separable case, introduce slack variables $\xi_i \geq 0$ and penalise their sum using a cost parameter $C > 0$.

Part (a): Linearly separable observations

Step 1. The primal optimisation problem is

$$\boxed{\min_{w,b} \ \frac{1}{2}\|w\|^2 \quad \text{subject to} \quad y_i(w^\top x_i + b) \geq 1, \ i = 1, \ldots, n.}$$

The objective tries to maximise the margin, and the constraints require correct separation with at least unit functional margin.

Step 2. Introduce Lagrange multipliers $\alpha_i \geq 0$ for the inequality constraints. The Lagrangian is

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} w^\top w - \sum_{i=1}^{n} \alpha_i \{y_i(w^\top x_i + b) - 1\}.$$

Step 3. Differentiate with respect to the primal variables and set derivatives equal to zero:

$$\frac{\partial \mathcal{L}}{\partial w} = w - \sum_{i=1}^{n} \alpha_i y_i x_i = 0 \quad \Longrightarrow \quad w = \sum_{i=1}^{n} \alpha_i y_i x_i,$$

$$\frac{\partial \mathcal{L}}{\partial b} = -\sum_{i=1}^{n} \alpha_i y_i = 0 \quad \Longrightarrow \quad \sum_{i=1}^{n} \alpha_i y_i = 0.$$

Step 4. Substitute $w = \sum_i \alpha_i y_i x_i$ into the Lagrangian. The dual problem becomes

$$\boxed{\max_{\alpha_1, \ldots, \alpha_n} \left\{ \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j x_i^\top x_j \right\}}$$

subject to

$$\boxed{\alpha_i \geq 0, \qquad \sum_{i=1}^{n} \alpha_i y_i = 0.}$$

Step 5. Suppose the solution α^* is known. Recover

$$w^* = \sum_{i=1}^{n} \alpha_i^* y_i x_i.$$

For a support vector x_{i_0} satisfying $\alpha_{i_0}^* > 0$, the margin equality gives

$$y_{i_0}(w^{*\top} x_{i_0} + b^*) = 1,$$

so

$$b^* = y_{i_0} - w^{*\top} x_{i_0}.$$

In practice, average this value over support vectors on the margin. A new point x is classified by

$$\boxed{\widehat{y} = \operatorname{sign}\left(\sum_{i=1}^{n} \alpha_i^* y_i x_i^\top x + b^* \right).}$$

Part (b): Non-linearly separable observations

Step 6. Introduce slack variables $\xi_i \geq 0$ and a penalty parameter $C > 0$. The primal problem becomes

$$\boxed{\min_{w, b, \xi} \ \frac{1}{2}\|w\|^2 + C \sum_{i=1}^{n} \xi_i}$$

subject to

$$y_i(w^\top x_i + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0.$$

A positive slack permits an observation to lie inside the margin or to be misclassified. The parameter C controls the trade-off between margin width and violations.

Step 7. The Lagrangian now has two sets of multipliers, $\alpha_i \geq 0$ for the margin constraints and $\mu_i \geq 0$ for $\xi_i \geq 0$:

$$\mathcal{L} = \frac{1}{2}\|w\|^2 + C\sum_i \xi_i - \sum_i \alpha_i\{y_i(w^\top x_i + b) - 1 + \xi_i\} - \sum_i \mu_i \xi_i.$$

Stationarity with respect to w, b, ξ_i gives

$$w = \sum_i \alpha_i y_i x_i, \qquad \sum_i \alpha_i y_i = 0, \qquad C - \alpha_i - \mu_i = 0.$$

Since $\mu_i \geq 0$, the last relation yields $0 \leq \alpha_i \leq C$.

Step 8. The dual objective has the same quadratic form as in the separable case, but the multipliers are bounded above:

$$\boxed{\max_\alpha \left\{ \sum_i \alpha_i - \frac{1}{2}\sum_i \sum_j \alpha_i \alpha_j y_i y_j x_i^\top x_j \right\}, \quad \sum_i \alpha_i y_i = 0, \quad 0 \leq \alpha_i \leq C.}$$

The future-observation classifier has the same expression, with the optimal soft-margin coefficients and intercept.

Final conclusion.
The hard-margin SVM minimises $\frac{1}{2}\|w\|^2$ subject to unit-margin separation, and its dual has $\alpha_i \geq 0$. The soft-margin SVM adds slack penalties, producing the same dual objective with the box constraints $0 \leq \alpha_i \leq C$. In both cases, classify a new point by the sign of $\sum_i \alpha_i^* y_i x_i^\top x + b^*$.

Question 67

Whole question from the assignment.

Let $x = (x_1, x_2)^\top$ and $y = (y_1, y_2)^\top$ be arbitrary observations in $\mathbb{R}^2$. For each of the following functions $K_i : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$, determine whether K_i is a reproducing kernel of some Hilbert space of functions on $\mathbb{R}^2$. Justify your answers.

1. $K_1(x, y) = x_1 y_1 + x_2 y_2$.
2. $K_2(x, y) = x_1^2 y_1^2 + 2x_1 x_2 y_1 y_2 + x_2^2 y_2^2$.
3. $K_3(x, y) = x_1^2 y_1^2 + x_2 y_2$.
4. $K_4(x, y) = 1 + x_1 y_1 + x_2 y_2$.
5. $K_5(x, y) = x_1 y_1 + x_2 y_2 + x_1 x_2 y_1 y_2$.
6. $K_6(x, y) = x_1^2 y_1^2 - x_2^2 y_2^2$.
7. $K_7(x, y) = x_1 y_1 + x_2 y_2 + x_1 y_2$.
8. $K_8(x, y) = \sin x_1 \sin y_1 + \cos x_1 \cos y_1 + \sin x_2 \sin y_2 + \cos x_2 \cos y_2$.
9. $K_9(x, y) = e^{x_1 + y_1} + e^{x_2 + y_2}$.
10. $K_{10}(x, y) = x_1^2 y_1^2 + x_2^2 y_2^2 + 1$.

Theory required for this solution.
A reproducing kernel must be symmetric and positive semidefinite. Positive semidefiniteness

means that for every finite set $x_1, \dots, x_m$ and real coefficients $c_1, \dots, c_m$,

$$\sum_{i=1}^{m} \sum_{j=1}^{m} c_i c_j K(x_i, x_j) \geq 0.$$

A convenient sufficient construction is a feature map ϕ such that

$$K(x, y) = \langle \phi(x), \phi(y) \rangle.$$

Then the Gram matrix is a matrix of inner products and is automatically positive semidefinite. A kernel that is not symmetric or that has $K(x, x) < 0$ for some x cannot be a reproducing kernel.

Step 1. $K_1(x, y) = x_1 y_1 + x_2 y_2$. Take $\phi_1(x) = (x_1, x_2)^\top$. Then $K_1 = \phi_1(x)^\top \phi_1(y)$. Hence **yes**.

Step 2. $K_2(x, y) = x_1^2 y_1^2 + 2 x_1 x_2 y_1 y_2 + x_2^2 y_2^2$. Take

$$\phi_2(x) = (x_1^2, \sqrt{2} x_1 x_2, x_2^2)^\top.$$

Its inner product with $\phi_2(y)$ is exactly $K_2(x, y)$. Hence **yes**.

Step 3. $K_3(x, y) = x_1^2 y_1^2 + x_2 y_2$. Take $\phi_3(x) = (x_1^2, x_2)^\top$. Then $K_3 = \phi_3(x)^\top \phi_3(y)$. Hence **yes**.

Step 4. $K_4(x, y) = 1 + x_1 y_1 + x_2 y_2$. Take $\phi_4(x) = (1, x_1, x_2)^\top$. Hence **yes**.

Step 5. $K_5(x, y) = x_1 y_1 + x_2 y_2 + x_1 x_2 y_1 y_2$. Take $\phi_5(x) = (x_1, x_2, x_1 x_2)^\top$. Hence **yes**.

Step 6. $K_6(x, y) = x_1^2 y_1^2 - x_2^2 y_2^2$. At $x = (0, 1)^\top$,

$$K_6(x, x) = 0 - 1 = -1 < 0.$$

A positive semidefinite kernel must have $K(x, x) \geq 0$ for every x. Hence **no**.

Step 7. $K_7(x, y) = x_1 y_1 + x_2 y_2 + x_1 y_2$. A reproducing kernel must be symmetric. Choose $x = (1, 0)^\top$ and $y = (0, 1)^\top$. Then

$$K_7(x, y) = 1, \qquad K_7(y, x) = 0.$$

It is not symmetric. Hence **no**.

Step 8. For K_8, take

$$\phi_8(x) = (\sin x_1, \cos x_1, \sin x_2, \cos x_2)^\top.$$

Then $K_8(x, y) = \phi_8(x)^\top \phi_8(y)$. Hence **yes**.

Step 9. For $K_9(x, y) = e^{x_1 + y_1} + e^{x_2 + y_2}$, take

$$\phi_9(x) = (e^{x_1}, e^{x_2})^\top.$$

Then $K_9(x, y) = \phi_9(x)^\top \phi_9(y)$. Hence **yes**.

Step 10. For $K_{10}(x, y) = x_1^2 y_1^2 + x_2^2 y_2^2 + 1$, take

$$\phi_{10}(x) = (1, x_1^2, x_2^2)^\top.$$

Then $K_{10}(x, y) = \phi_{10}(x)^\top \phi_{10}(y)$. Hence **yes**.

Final conclusion.

Kernel	K_1	K_2	K_3	K_4	K_5	K_6	K_7	K_8	K_9	K_{10}
RKHS kernel?	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes

The positive cases have explicit finite-dimensional feature maps. K_6 fails non-negativity on the diagonal, and K_7 fails symmetry.

Question 68

Whole question from the assignment.

Suppose we are given a binary classification problem with training observations (x_i, y_i), $i = 1, \ldots, n$, where $x_i \in \mathbb{R}^2$ and $y_i \in \{-1, +1\}$. The two classes are not linearly separable in the original input space.

1. Explain how a nonlinear SVM can be constructed using a feature map $\phi : \mathbb{R}^2 \to \mathcal{H}$, where $\mathcal{H}$ is a higher-dimensional feature space.
2. Derive the dual optimization problem for a soft-margin SVM using a kernel function $K(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle_{\mathcal{H}}$.
3. Consider the Gaussian radial basis function (RBF) kernel

$$K(x, z) = \exp(-\gamma \|x - z\|^2), \qquad \gamma > 0.$$

   Explain why this kernel allows nonlinear decision boundaries.
4. Given the optimal dual coefficients $\alpha_1^*, \ldots, \alpha_n^*$ and intercept b^*, write down the decision function for classifying a new observation x.
5. Explain how the support vectors are identified from the optimal dual coefficients. Distinguish between support vectors lying on the margin and those lying inside the margin or being misclassified.

Theory required for this solution.
A feature map transforms the original observation into a vector $\phi(x)$ in a Hilbert space. A linear separator in that feature space can correspond to a nonlinear boundary in the original input space. The kernel trick replaces explicit inner products by a kernel $K(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle$. For non-separable data, use soft-margin slack variables and the penalty parameter C.

Step 1. Map each input to $\phi(x_i) \in \mathcal{H}$ and seek a hyperplane

$$\langle w, \phi(x) \rangle_{\mathcal{H}} + b = 0.$$

The soft-margin primal problem is

$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|_{\mathcal{H}}^2 + C \sum_{i=1}^{n} \xi_i$$

subject to

$$y_i\{\langle w, \phi(x_i) \rangle_{\mathcal{H}} + b\} \geq 1 - \xi_i, \qquad \xi_i \geq 0.$$

Step 2. Introduce multipliers $\alpha_i \geq 0$ and $\mu_i \geq 0$. The Lagrangian is

$$\begin{aligned}\mathcal{L} = \frac{1}{2}\|w\|_{\mathcal{H}}^2 + C\sum_i \xi_i \\ - \sum_i \alpha_i \left[y_i\{\langle w, \phi(x_i)\rangle + b\} - 1 + \xi_i\right] - \sum_i \mu_i \xi_i.\end{aligned}$$

Stationarity gives

$$w = \sum_i \alpha_i y_i \phi(x_i), \qquad \sum_i \alpha_i y_i = 0, \qquad C - \alpha_i - \mu_i = 0.$$

Since $\mu_i \geq 0$, we obtain $0 \leq \alpha_i \leq C$.

Step 3. Substitute the expression for w into the Lagrangian. Every inner product becomes a kernel evaluation:

$$\langle \phi(x_i), \phi(x_j)\rangle_{\mathcal{H}} = K(x_i, x_j).$$

The dual problem is

$$\boxed{\max_{\alpha}\left\{\sum_{i=1}^n \alpha_i - \frac{1}{2}\sum_{i=1}^n\sum_{j=1}^n \alpha_i\alpha_j y_i y_j K(x_i, x_j)\right\}}$$

subject to

$$\boxed{\sum_{i=1}^n \alpha_i y_i = 0, \qquad 0 \leq \alpha_i \leq C.}$$

Step 4. The Gaussian RBF kernel depends on squared distance:

$$K(x, z) = e^{-\gamma\|x-z\|^2}.$$

It is a positive definite kernel and corresponds to a feature representation with many nonlinear features (in fact, an infinite-dimensional feature space). A hyperplane in that feature space therefore pulls back to a generally nonlinear boundary in the original two-dimensional plane. The parameter γ controls how rapidly similarity decreases with distance.

Step 5. From stationarity,

$$w^* = \sum_{i=1}^n \alpha_i^* y_i \phi(x_i).$$

For a new observation x,

$$\begin{aligned}\langle w^*, \phi(x)\rangle + b^* &= \sum_{i=1}^n \alpha_i^* y_i \langle \phi(x_i), \phi(x)\rangle + b^* \\ &= \sum_{i=1}^n \alpha_i^* y_i K(x_i, x) + b^*.\end{aligned}$$

Thus

$$\boxed{\widehat{y} = \operatorname{sign}\left\{\sum_{i=1}^n \alpha_i^* y_i K(x_i, x) + b^*\right\}.}$$

Step 6. The support vectors are precisely the training observations with $\alpha_i^* > 0$. The KKT complementary-slackness conditions distinguish their roles:

$$\alpha_i\{1 - \xi_i - y_i(\langle w, \phi(x_i)\rangle + b)\} = 0, \qquad \mu_i\xi_i = 0.$$

If $0 < \alpha_i^* < C$, then $\mu_i^* = C - \alpha_i^* > 0$, so $\xi_i^* = 0$, and the point lies exactly on the margin:

$$y_i(\langle w^*, \phi(x_i)\rangle + b^*) = 1.$$

If $\alpha_i^* = C$, the point may lie inside the margin ($0 < \xi_i < 1$), on the wrong side but not beyond the margin error threshold ($\xi_i = 1$), or be misclassified ($\xi_i > 1$). A point with $\alpha_i^* = 0$ is not a support vector and is outside the active margin constraints in the usual non-degenerate case.

Final conclusion.

The soft-margin kernel SVM dual is

$$\boxed{\max_{\alpha}\left[\sum_i \alpha_i - \frac{1}{2}\sum_{i,j}\alpha_i\alpha_j y_i y_j K(x_i, x_j)\right]}$$

subject to $\sum_i \alpha_i y_i = 0$ and $0 \le \alpha_i \le C$. The RBF kernel makes the boundary nonlinear in the input plane. The decision function is the kernel expansion above; support vectors have $\alpha_i^* > 0$, with $0 < \alpha_i^* < C$ identifying margin points and $\alpha_i^* = C$ allowing margin violations or misclassification.

Question 69

Whole question from the assignment.

> For the classification problems in Question no. 30 and Question no. 31, what type of kernel will you recommend for classification using nonlinear SVM? Justify your answer.

Theory required for this solution.

A kernel SVM replaces ordinary inner products by a positive semidefinite kernel. A polynomial kernel of degree 2 introduces quadratic features and pairwise interactions. In particular,

$$K(x, z) = (1 + x^\top z)^2$$

contains constant, linear, square, and interaction terms. A nonlinear SVM can therefore learn a quadratic boundary in the original coordinates.

Step 1. Identify the geometry in Question 30. The two classes are diagonal mixtures: one has component means $(1, 1)$ and $(-1, -1)$, and the other has means $(-1, 1)$ and $(1, -1)$. Comparing their mixture densities shows that the Bayes rule is determined by the sign of x_1x_2:

$$\text{Class 1 if } x_1x_2 > 0, \qquad \text{Class 2 if } x_1x_2 < 0.$$

The separating boundary is the pair of coordinate axes, a quadratic boundary expressed by $x_1x_2 = 0$.

Step 2. Identify the geometry in Question 31. In the transformed class, the two coordinates have the same sign: the transformation maps all four original Gaussian quadrants into the two same-sign quadrants. Its density is twice the standard bivariate normal density on those quadrants and zero on the opposite-sign quadrants. Against the full standard normal class with equal priors, the Bayes rule again separates same-sign from opposite-sign quadrants, with boundary $x_1x_2 = 0$.

Step 3. A degree-2 polynomial kernel is therefore structurally appropriate. For example,

$$K(x, z) = (1 + x^\top z)^2$$

corresponds to a feature map containing x_1^2, x_2^2, and x_1x_2 (along with constant and linear terms). This gives the SVM the interaction feature needed to represent the quadratic sign pattern. A homogeneous quadratic kernel can also be used if one deliberately wants only quadratic terms.

Step 4. A Gaussian RBF kernel is a flexible alternative if the exact form of the boundary is not known or if a smoother and more general nonlinear boundary is desired. However, based on the known form of these two problems, degree 2 is the more directly motivated choice.

Final conclusion.

Recommend a **degree-2 polynomial kernel**, such as

$$\boxed{K(x,z) = (1 + x^\top z)^2,}$$

because both questions have a Bayes boundary determined by $x_1x_2 = 0$. The RBF kernel is a reasonable flexible alternative, but the quadratic polynomial kernel directly matches the interaction structure of the two classification problems.

Question 70

Whole question from the assignment.

> What is the main idea of boosting? What are its advantages and disadvantages over bagging? Describe the AdaBoost and Gradient Boosting algorithms for regression and classification. What are the main differences between Gradient Boosting and XGBoost?

Theory required for this solution.

Boosting constructs an ensemble sequentially. Later learners focus on what the current ensemble has not yet explained well. In the supplied W15 notes, bagging is described as a variance-reduction procedure with equal observation weights and parallel fitting, while boosting adjusts observation weights sequentially and aggregates learners using learner-specific weights. The notes give the discrete AdaBoost classification algorithm, including its weighted error, learner weight, and observation-weight update.

Step 1. Main idea and comparison with bagging. Bagging draws many bootstrap samples, fits learners independently, and averages/votes. Boosting instead adds learners one at a time. It increases emphasis on observations that are currently difficult to classify or predict. The main distinctions are:

	Bagging	Boosting
Primary effect	Mainly reduces variance	Can reduce bias as well as variance
Observation weights	Usually equal	Updated from one stage to the next
Fitting	Learners can be fitted in parallel	Sequential; later learners depend on earlier ones
Aggregation	Average or majority vote	Weighted/additive combination
Sensitivity	Bootstrap averaging stabilises individual trees	Difficult observations receive increased emphasis

Step 2. AdaBoost for binary classification (the version explicitly developed in W15). Let

$y_i \in \{-1, +1\}$. Initialise

$$F_0(x) = 0, \qquad W_0(i) = \frac{1}{n}.$$

At boosting step m, fit a weak classifier $\delta_m(x) \in \{-1, +1\}$ using the current observation weights. Compute its weighted error

$$\epsilon_m = \frac{\sum_{i=1}^n W_{m-1}(i) I\{y_i \neq \delta_m(x_i)\}}{\sum_{i=1}^n W_{m-1}(i)}.$$

Require $\epsilon_m < 1/2$ and set

$$\alpha_m = \frac{1}{2} \log \frac{1 - \epsilon_m}{\epsilon_m}.$$

Update the additive score and observation weights:

$$F_m(x) = F_{m-1}(x) + \alpha_m \delta_m(x),$$

$$W_m(i) = W_{m-1}(i) \exp\{-\alpha_m y_i \delta_m(x_i)\},$$

followed by normalisation so that the weights sum to one. Correctly classified observations are downweighted by $e^{-\alpha_m}$; misclassified observations are upweighted by e^{α_m}. The final classifier is

$$\boxed{\delta_{\text{Ada}}(x) = \text{sign}\left\{\sum_{m=1}^M \alpha_m \delta_m(x)\right\}.}$$

Step 3. AdaBoost for more than two classes. The supplied W15 notes also state the multiclass SAMME scheme. For K classes, calculate the weighted error ϵ_m and use a learner weight adjusted for the number of classes:

$$\alpha_m = \frac{1}{2} \log \frac{1 - \epsilon_m}{\epsilon_m} + \log(K - 1).$$

Misclassified observations are upweighted and correctly classified observations are downweighted. The final class is the one receiving the largest accumulated weighted vote:

$$\delta_{\text{SAMME}}(x) = \underset{1 \leq j \leq K}{\text{argmax}} \sum_{m=1}^M \alpha_m I\{\delta_m(x) = j\}.$$

Step 4. AdaBoost-style regression (one common variant, AdaBoost.R2; supplementary, not developed in W15). Regression needs a numerical measure of how badly each fitted value misses its response. Initialise $W_1(i) = 1/n$. At stage m, fit a regression learner f_m using the current observation weights, and compute the absolute residuals

$$e_{im} = |y_i - f_m(x_i)|.$$

If all residuals are zero, the current learner is already perfect on the training sample and the procedure can stop. Otherwise normalise them by the largest residual:

$$L_{im} = \frac{e_{im}}{\max_r e_{rm}}, \qquad 0 \leq L_{im} \leq 1.$$

The weighted average loss is

$$E_m = \sum_{i=1}^n W_m(i) L_{im}.$$

If $E_m \geq 1/2$, reject the learner and stop (or refit with a better learner). Otherwise define

$$\beta_m = \frac{E_m}{1 - E_m}, \qquad \alpha_m = \log \frac{1}{\beta_m}.$$

Update the observation weights by

$$W_{m+1}(i) = W_m(i)\,\beta_m^{1-L_{im}},$$

and renormalise them to sum to one. A small loss gives a factor close to $\beta_m < 1$ and hence decreases that observation's relative weight; a large loss gives a factor closer to 1 and hence increases its relative weight compared with well-fitted observations. The final regression prediction is the weighted median of the learner predictions $f_m(x)$, using learner weights α_m. This is one standard AdaBoost.R2 construction; other regression variants use different loss and aggregation rules.

Step 5. Gradient Boosting for regression (supplementary algorithmic explanation). The supplied notes do not derive Gradient Boosting, but its general stagewise idea can be written directly as minimisation of an empirical loss. Start with a constant prediction F_0 that minimises the chosen loss. At stage m, compute the negative gradients, also called pseudo-residuals:

$$r_{im} = -\left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\right]_{F=F_{m-1}}.$$

Fit a weak regression learner $h_m(x)$ (often a shallow tree) to these pseudo-residuals, choose a step size ρ_m by line search or leaf-wise optimisation, and update

$$F_m(x) = F_{m-1}(x) + \nu\rho_m h_m(x),$$

where $0 < \nu \leq 1$ is a learning-rate or shrinkage parameter. For squared-error loss $L(y, F) = \frac{1}{2}(y-F)^2$, the pseudo-residual is simply

$$r_{im} = y_i - F_{m-1}(x_i),$$

so each new tree is fitted to the current residuals. The final regression prediction is $F_M(x)$.

Step 6. Gradient Boosting for classification (supplementary algorithmic explanation). For binary labels $y_i \in \{0, 1\}$, use a logistic loss and represent the ensemble by a score $F(x)$, with

$$p(x) = \frac{1}{1 + e^{-F(x)}}.$$

The negative gradient with respect to the current score is

$$r_{im} = y_i - p_{m-1}(x_i).$$

At each step, fit a regression learner to these pseudo-residuals and update the score by a shrunken fitted learner, with a line search or node-wise update chosen for the logistic loss. Predict Class 1 when the final fitted probability exceeds 1/2. For K classes, maintain K score functions and convert them to probabilities with the softmax transformation.

Step 7. Gradient Boosting versus XGBoost (supplementary explanation; not developed in the uploaded notes). Gradient Boosting is the general stagewise functional-gradient framework described above. XGBoost is a particular highly engineered implementation of gradient-boosted decision trees. Its commonly used objective adds an explicit complexity penalty on tree structure and leaf weights, and its split calculations use first- and second-order loss derivatives. It also includes implementation features such as shrinkage, row/column subsampling, regularisation, efficient split search, and handling of missing predictor values. Thus, XGBoost is not a different ensemble principle; it is a regularised and computationally optimised implementation of gradient-boosted trees.

Step 8. State the scope limitation clearly. W15 explicitly derives discrete AdaBoost for binary classification and gives its multiclass SAMME extension. The uploaded notes do not contain AdaBoost.R2, a Gradient Boosting derivation, or an XGBoost description. Therefore, the AdaBoost.R2, Gradient Boosting, and XGBoost portions above are labelled supplementary conceptual completions, not material quoted from the provided lecture notes. In particular, AdaBoost regression has multiple variants, so the solution has stated exactly which one is being described.

Final conclusion.

Boosting builds an additive ensemble sequentially, giving later learners emphasis on the remaining errors. Relative to bagging, it is sequential and can reduce bias as well as variance, but is less naturally parallel and can concentrate too strongly on noisy or difficult observations. W15's AdaBoost classifier uses weighted errors, learner weights α_m, and multiplicative observation-weight updates. Gradient Boosting generalises the stagewise idea by fitting learners to negative loss gradients. XGBoost is a regularised, second-order and computationally optimised gradient-boosted-tree implementation. AdaBoost.R2 and the Gradient Boosting/XGBoost descriptions are not part of the supplied lecture notes and have been identified as supplementary rather than attributed to them.